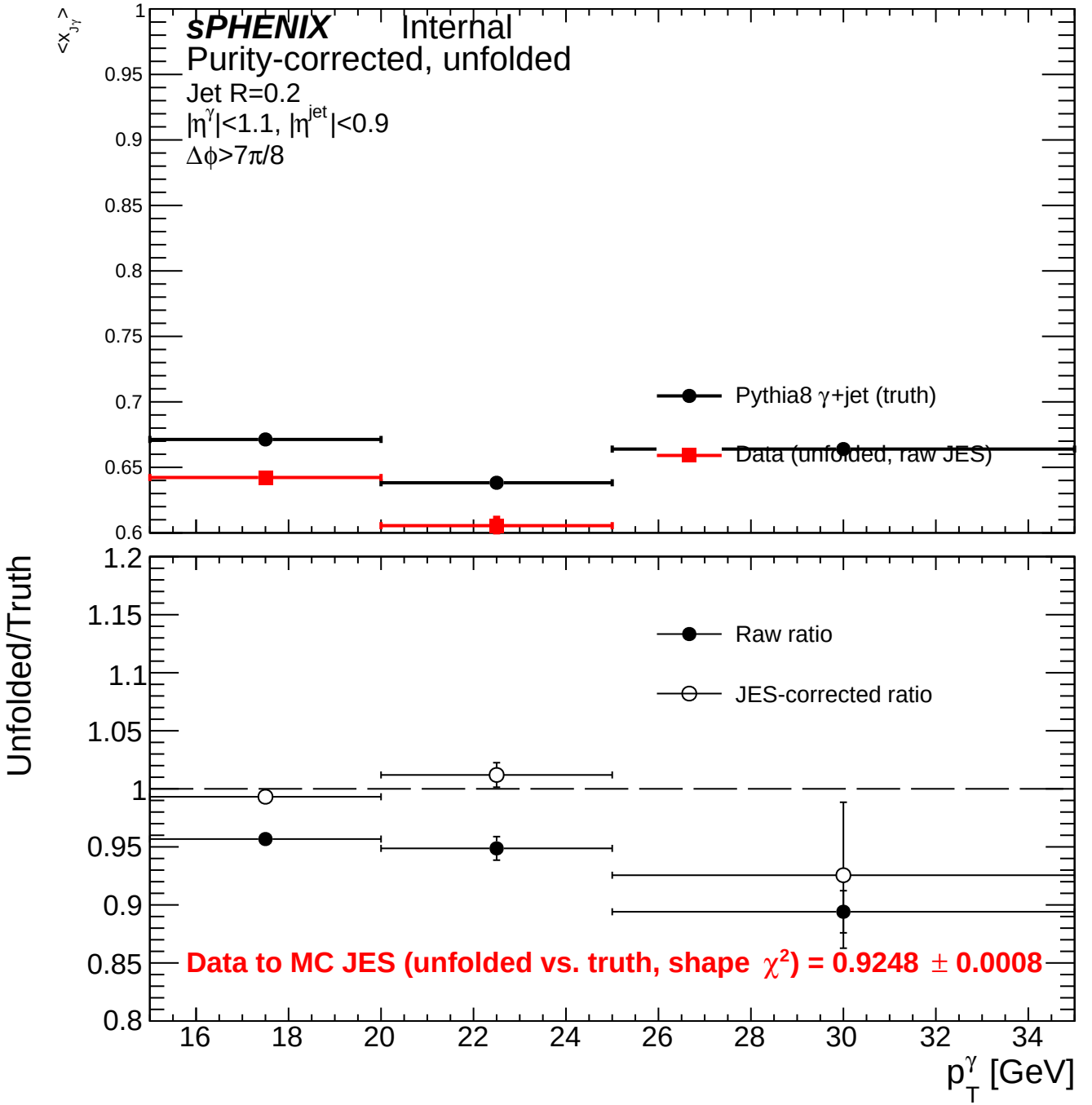
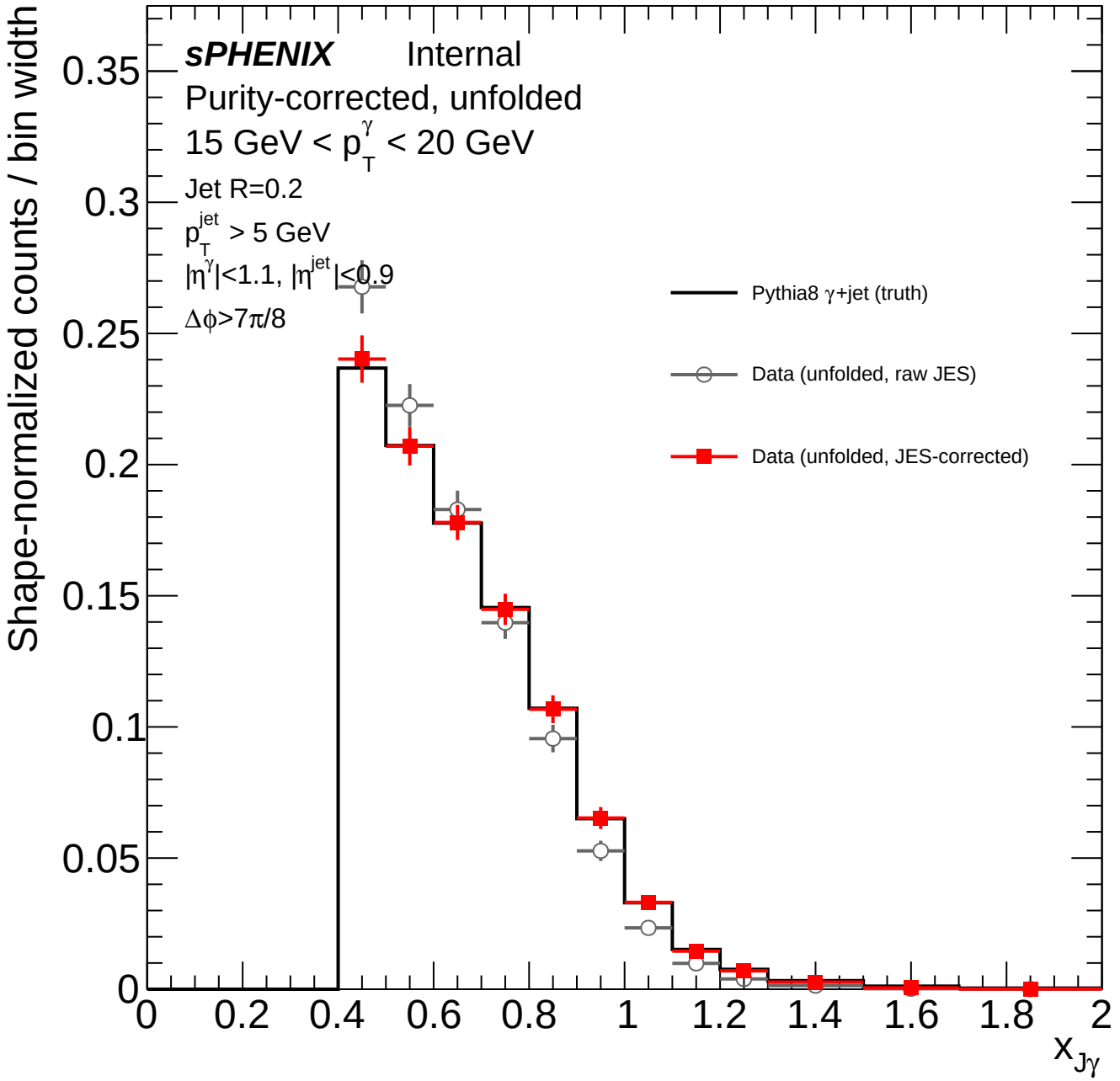
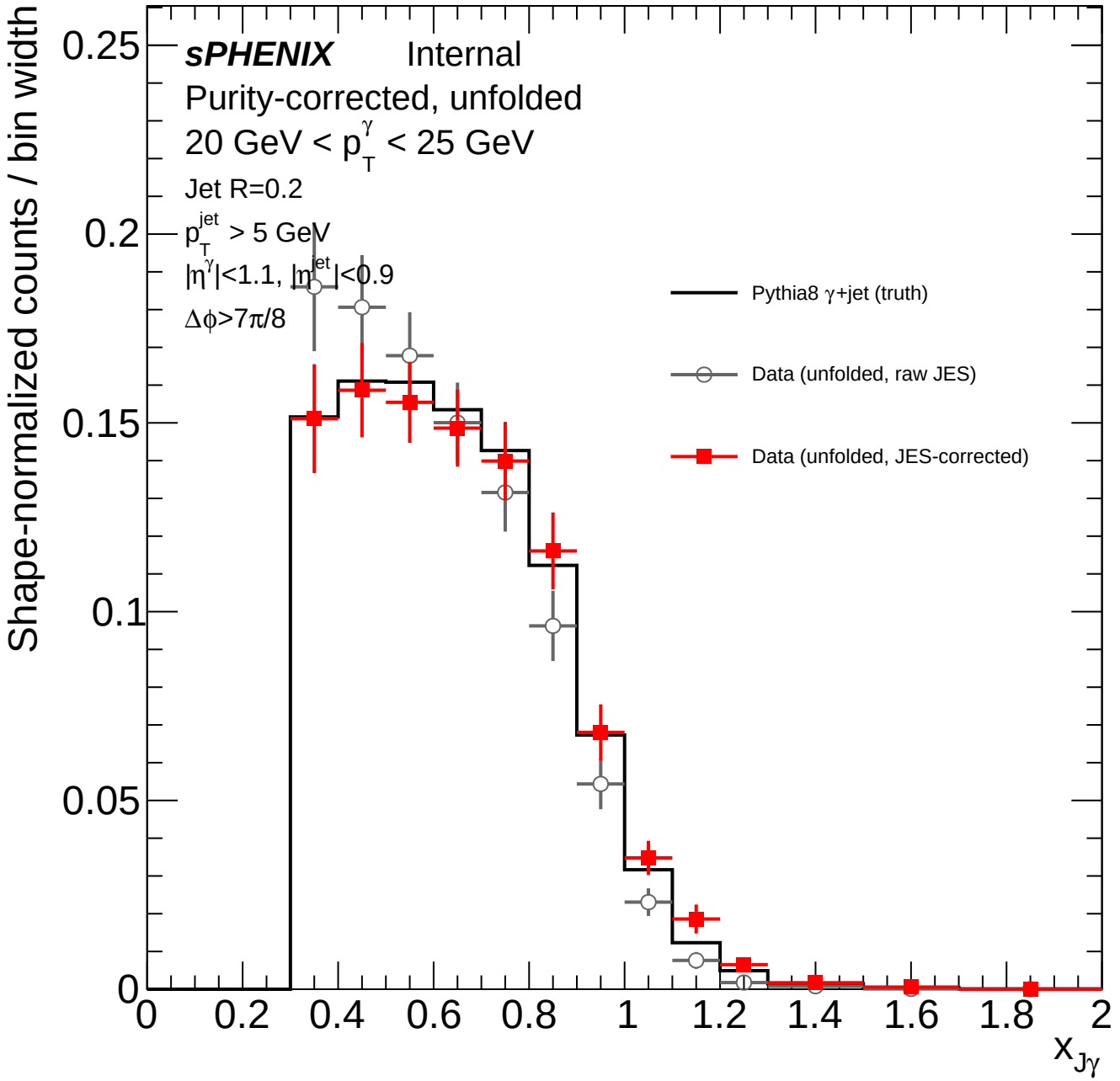
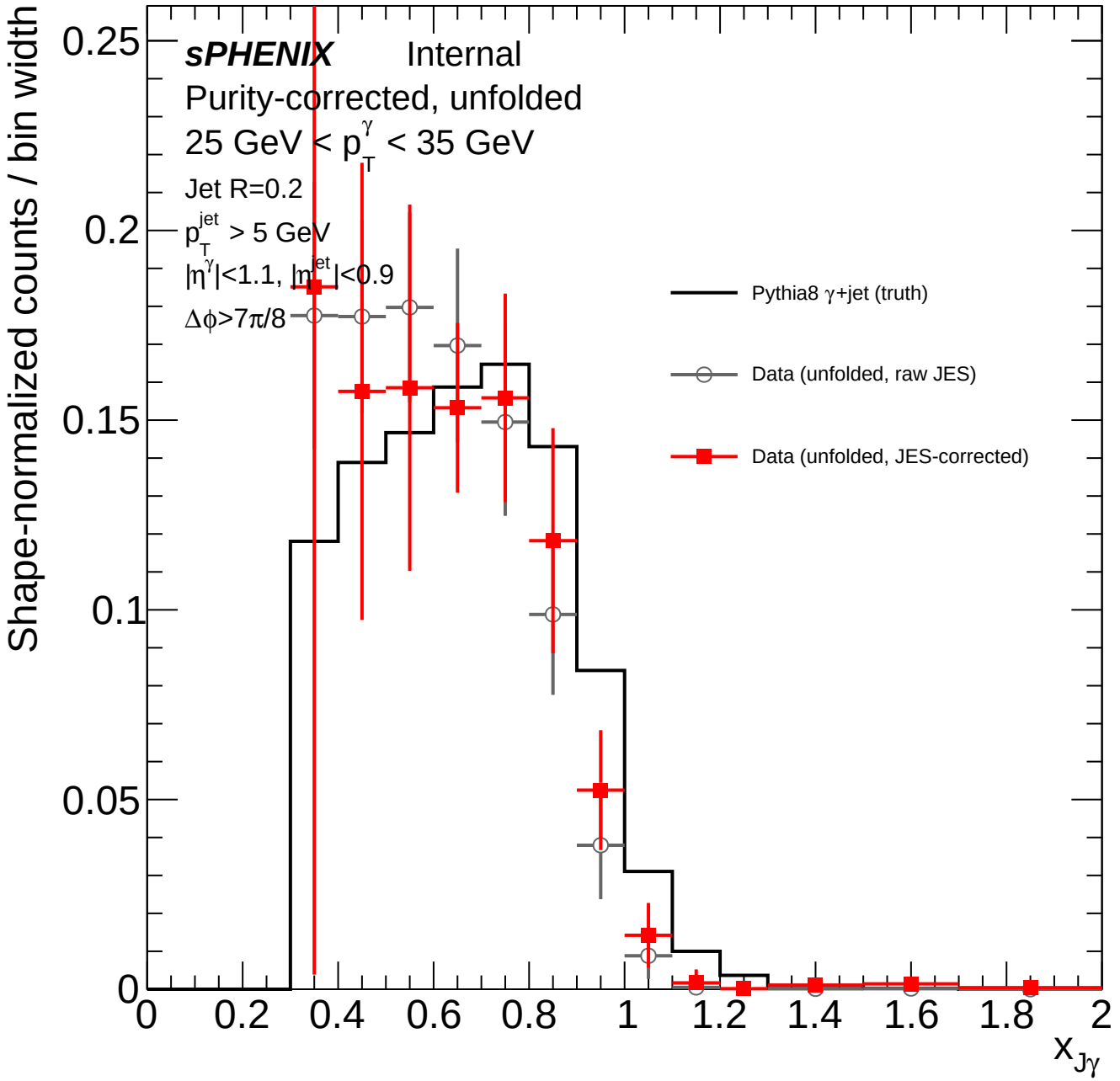
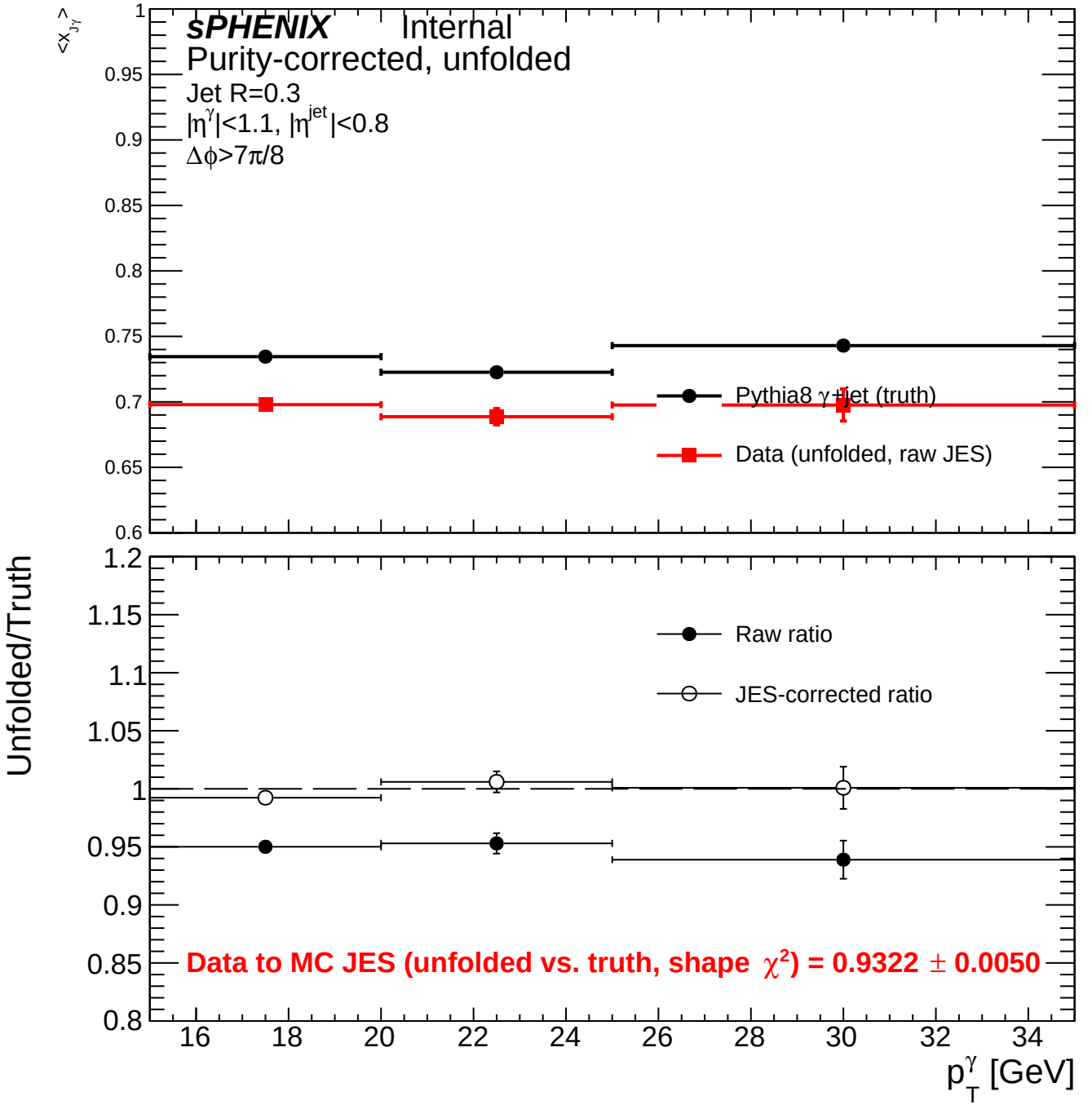
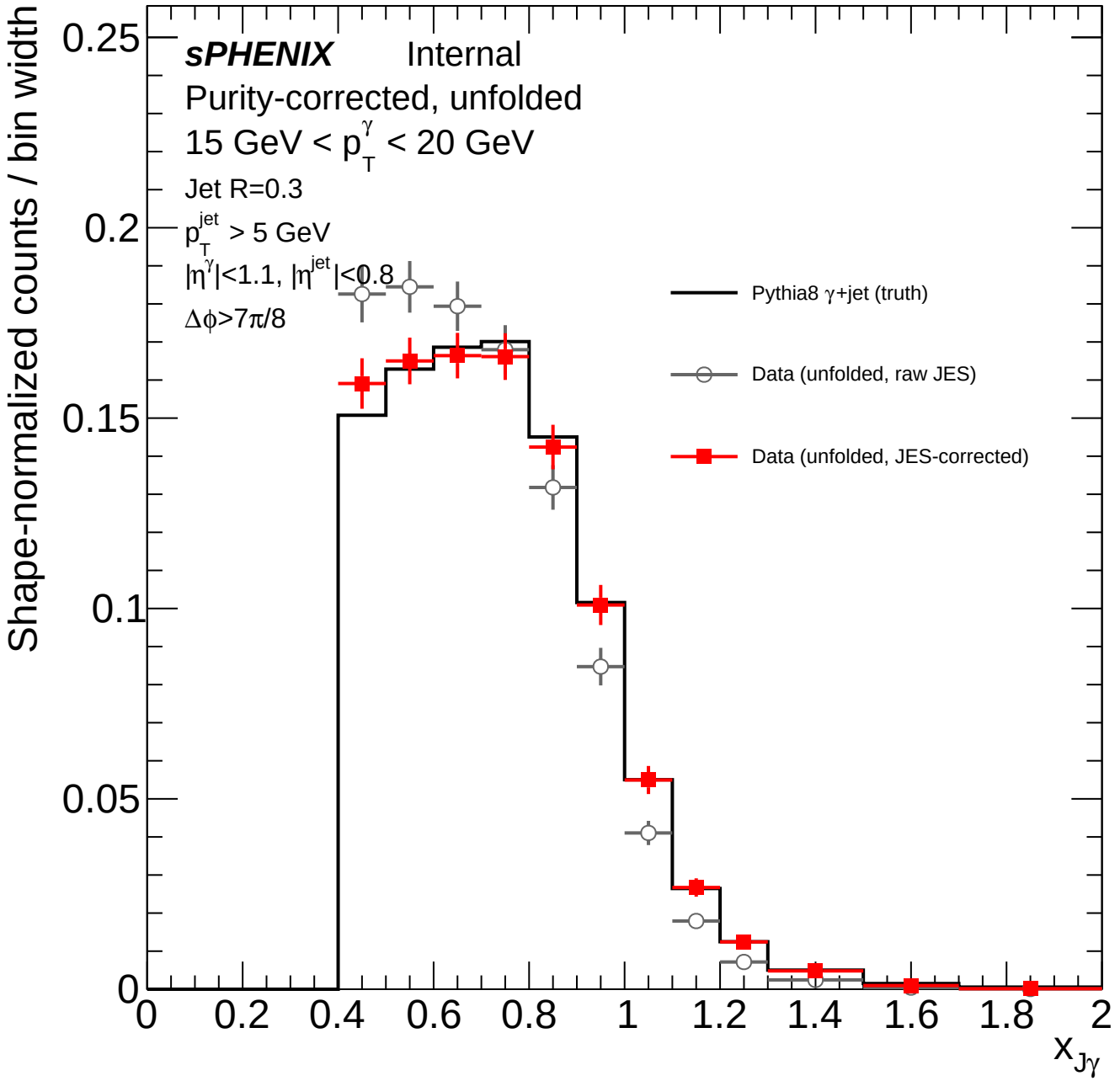

sPHENIX Internal
Purity-corrected, unfolded
Jet R=0.2
|η^γ|<1.1, |η^jet|<0.9
Δφ>7π/8
<x_jγ>
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Unfolded/Truth
Raw ratio
JES-corrected ratio
Data to MC JES (unfolded vs. truth, shape χ²) = 0.9248 ± 0.0008
p_T^γ [GeV]

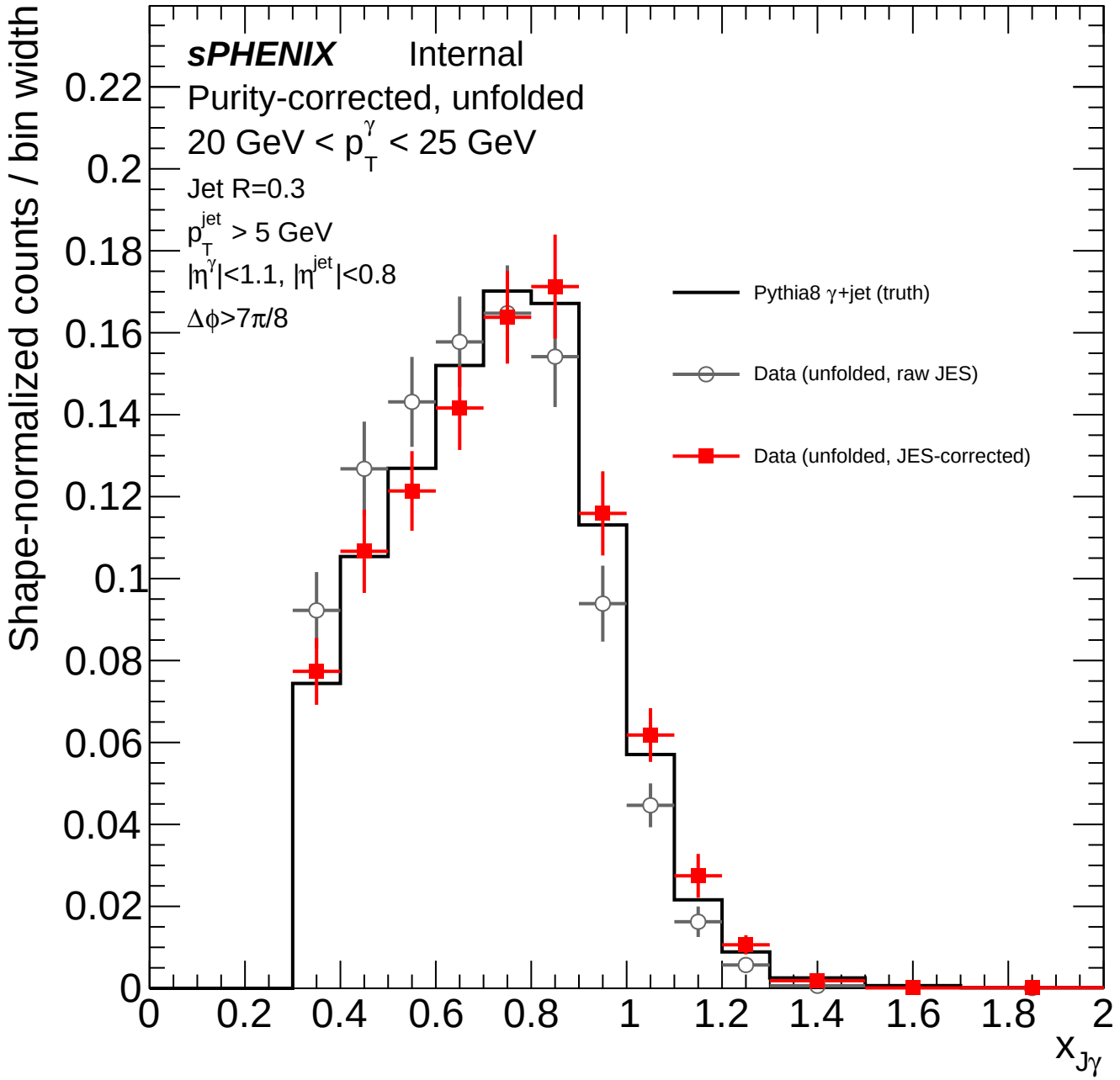

sPHENIX Internal
Purity-corrected, unfolded
15 GeV < p_T^γ < 20 GeV
Jet R=0.2
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.9
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
x_Jγ

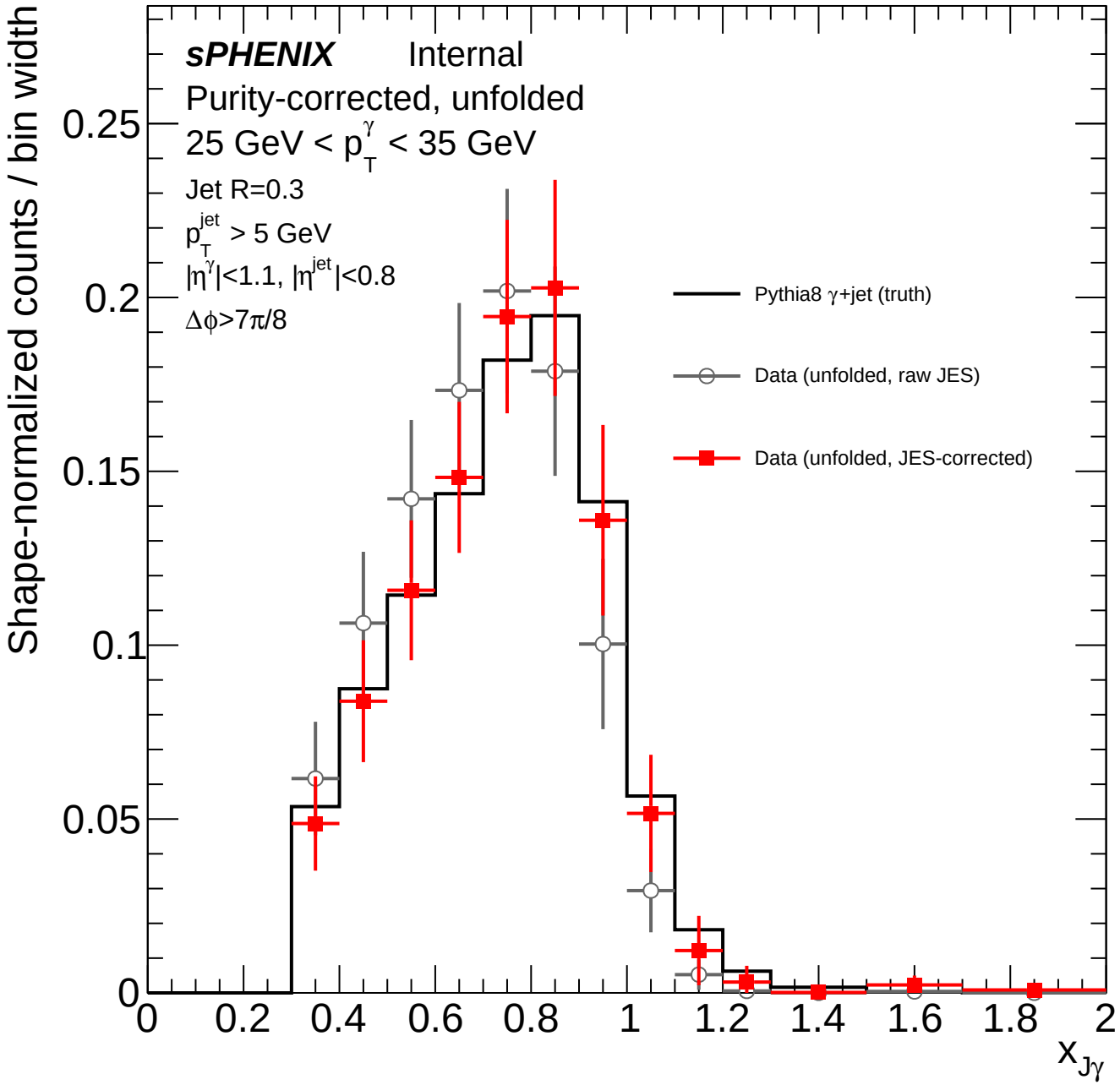

sPHENIX Internal
Purity-corrected, unfolded
20 GeV < p_T^γ < 25 GeV
Jet R=0.2
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.9
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

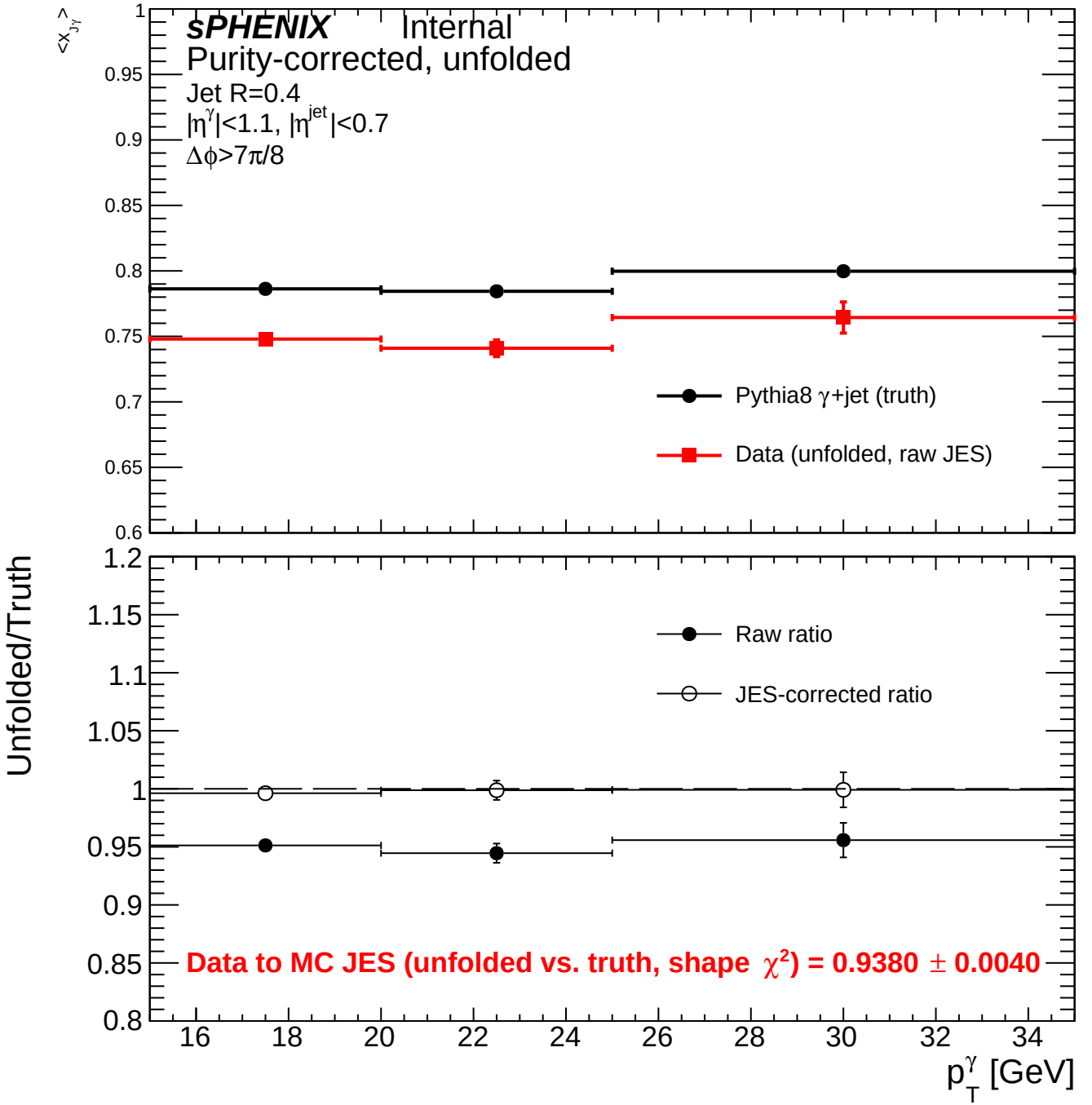

sPHENIX Internal
Purity-corrected, unfolded
25 GeV < p_T^γ < 35 GeV
Jet R=0.2
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.9
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

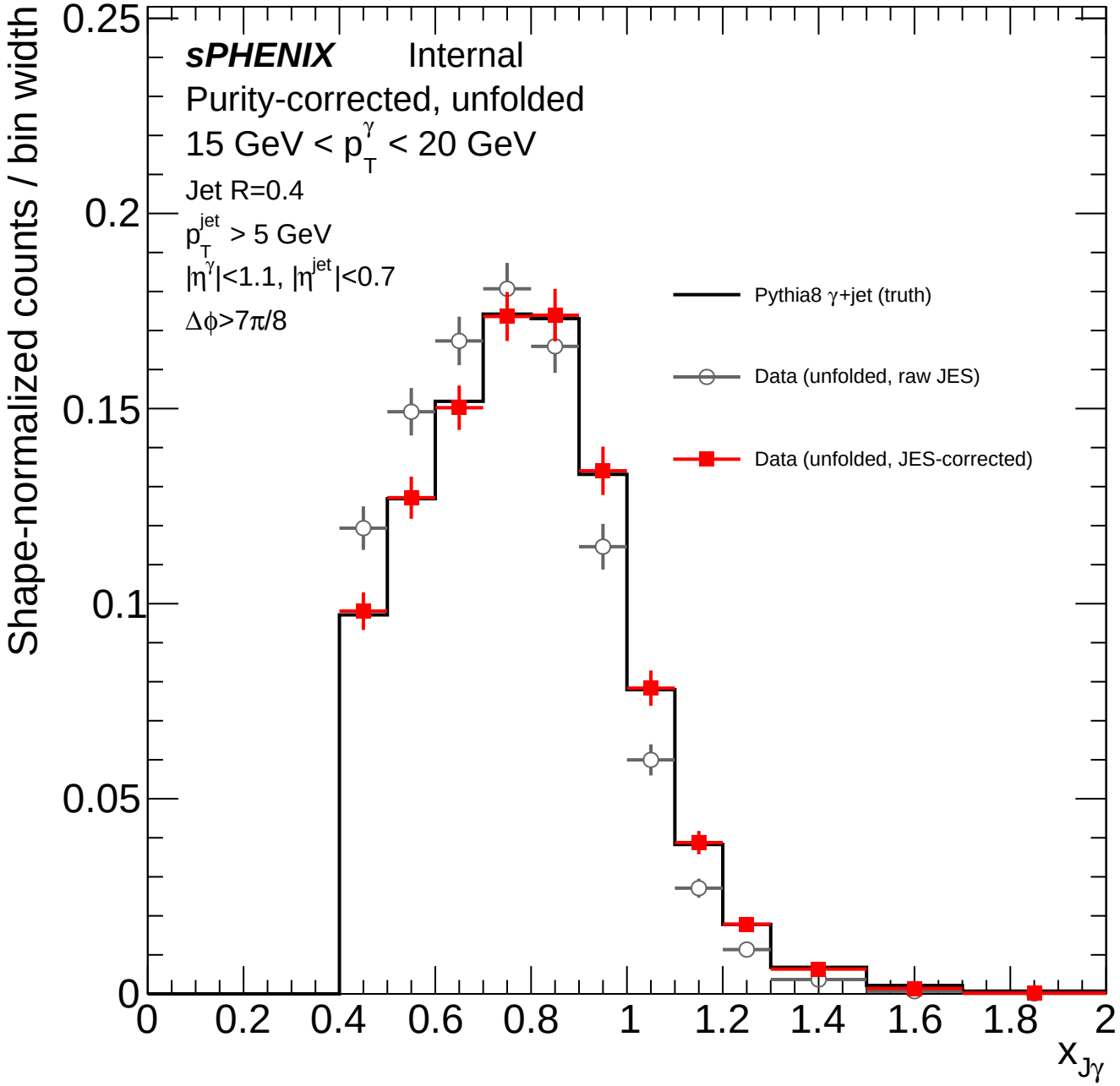

sPHENIX Internal
Purity-corrected, unfolded
Jet R=0.3
|η^γ|<1.1, |η^jet|<0.8
Δφ>7π/8
<x_jγ>
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Unfolded/Truth
Raw ratio
JES-corrected ratio
Data to MC JES (unfolded vs. truth, shape χ²) = 0.9322 ± 0.0050
p_T^γ [GeV]

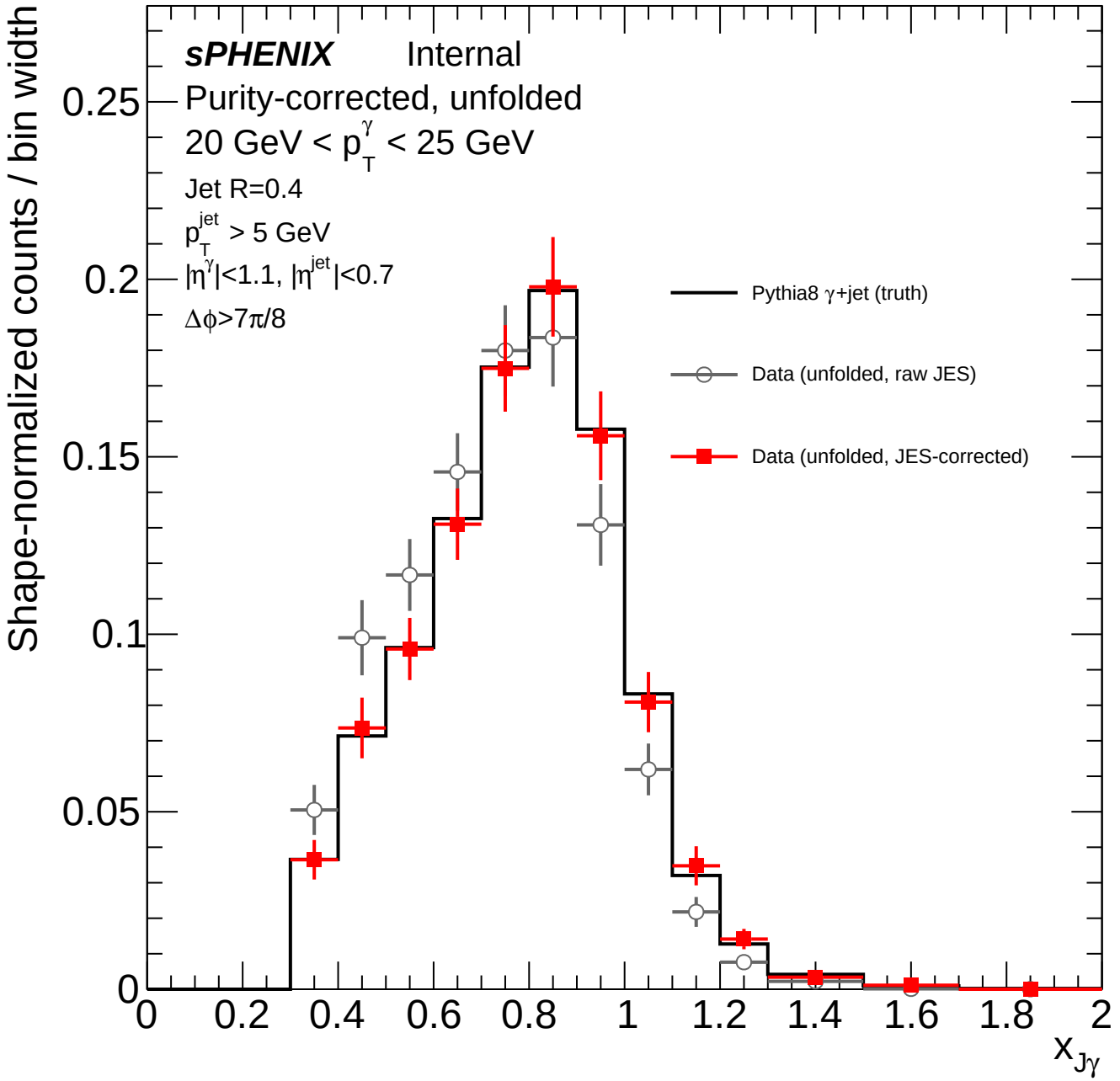

sPHENIX Internal
Purity-corrected, unfolded
15 GeV < p_T^γ < 20 GeV
Jet R=0.3
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.8
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

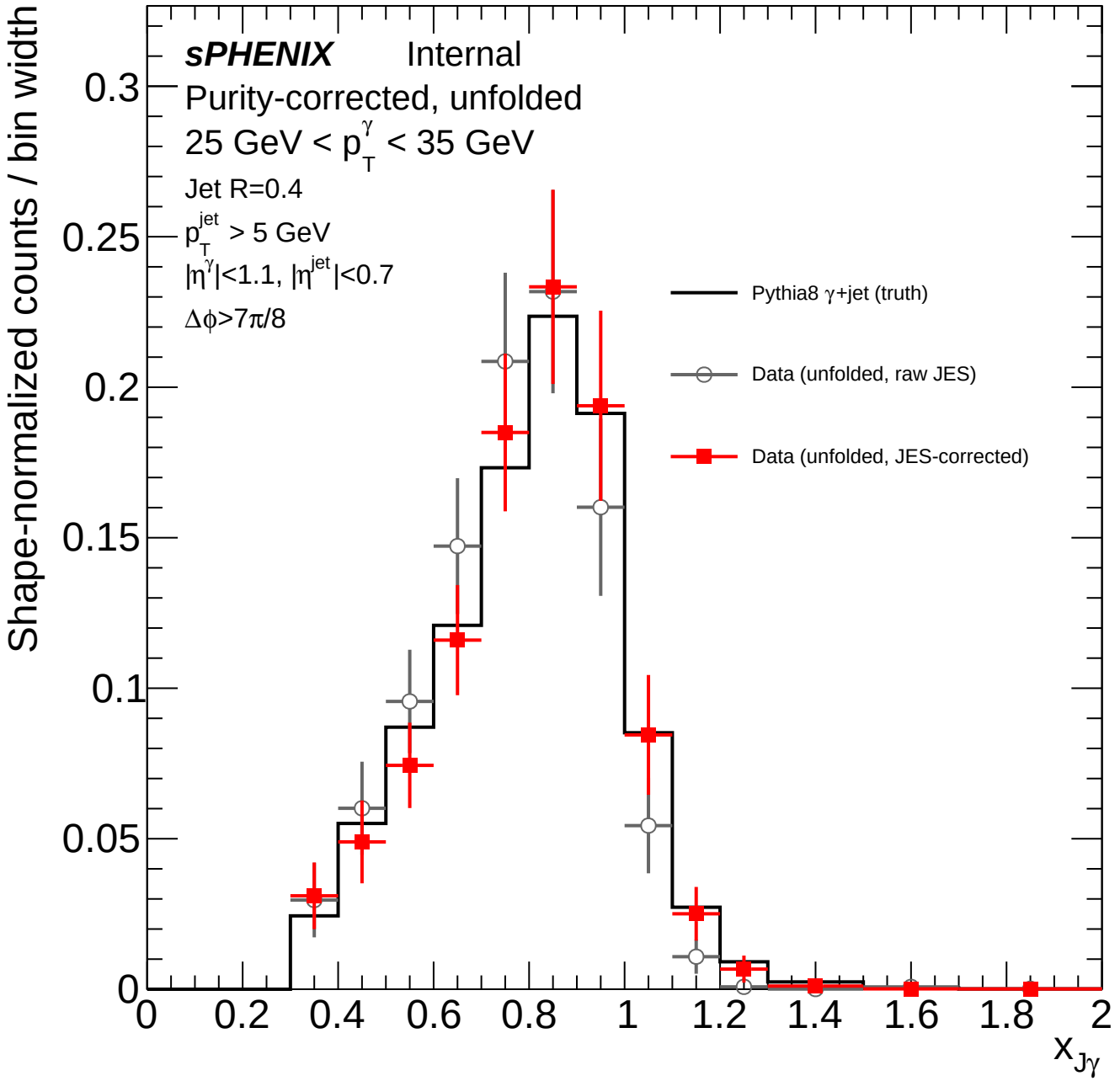

sPHENIX Internal
Purity-corrected, unfolded
20 GeV < p_T^γ < 25 GeV
Jet R=0.3
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.8
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
x_Jγ
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

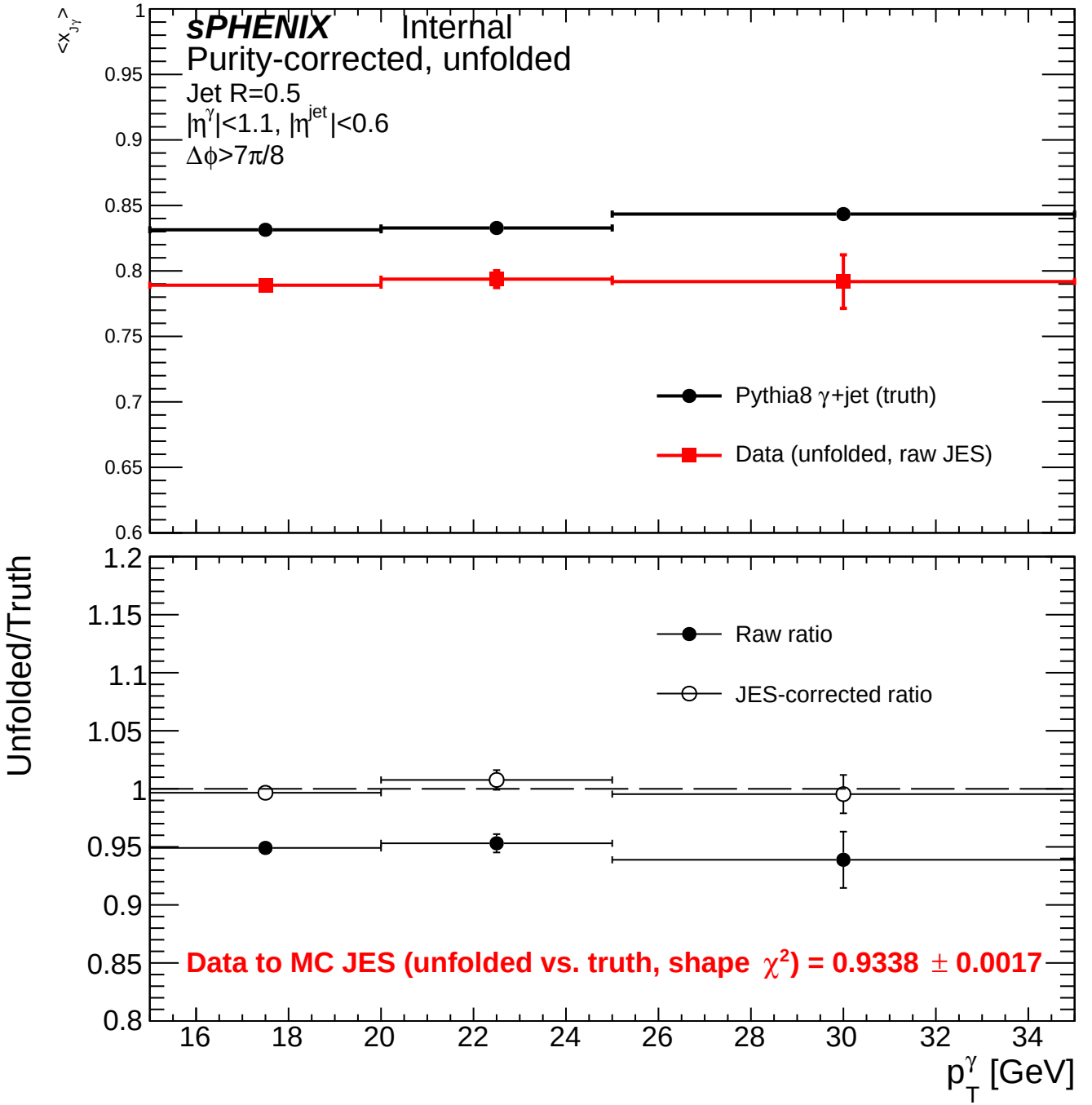

sPHENIX Internal
Purity-corrected, unfolded
25 GeV < p_T^γ < 35 GeV
Jet R=0.3
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.8
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
0
0.05
0.1
0.15
0.2
0.25
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
x_Jγ

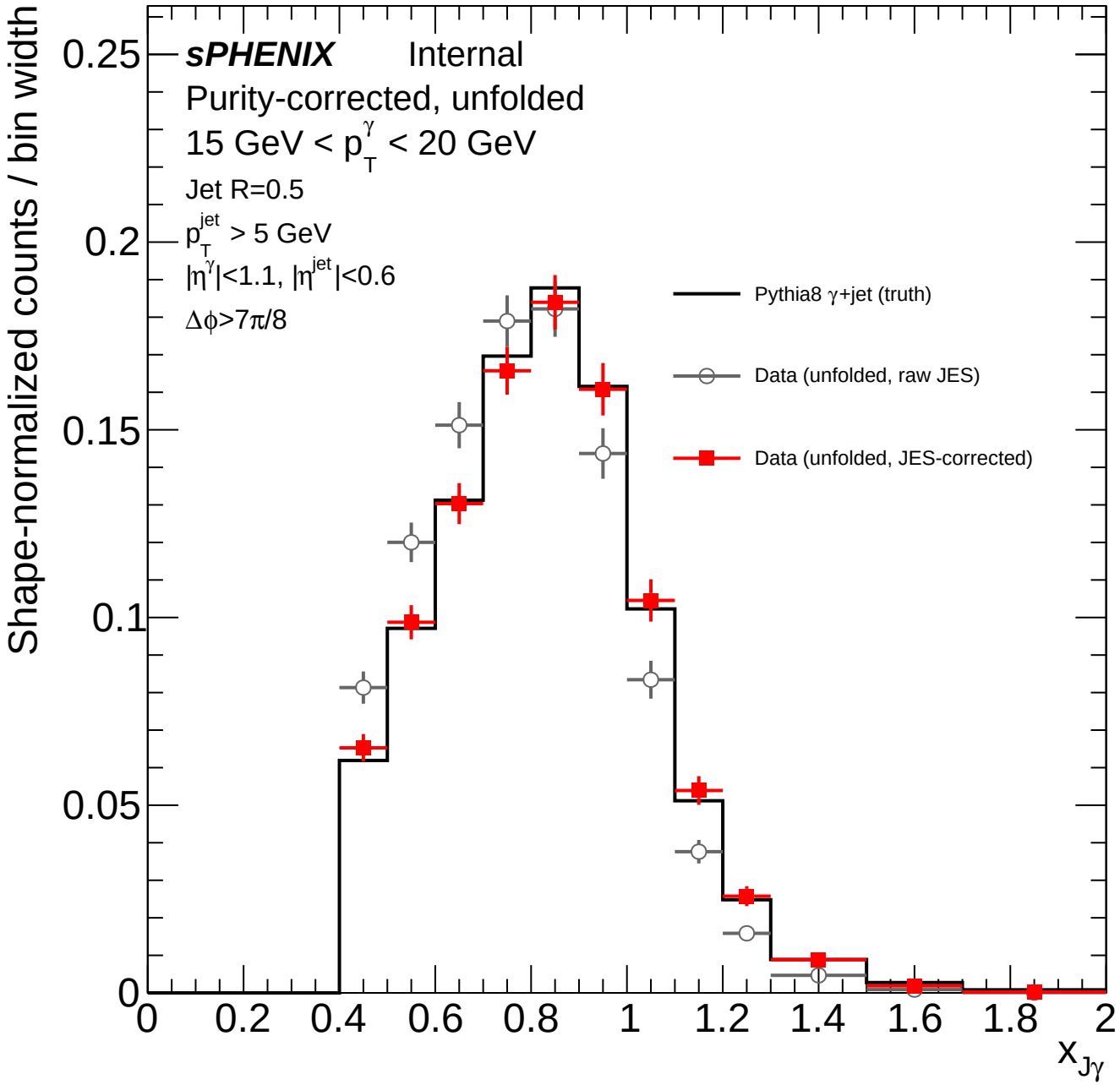

sPHENIX Internal
Purity-corrected, unfolded
Jet R=0.4
|η^γ|<1.1, |η^jet|<0.7
Δφ>7π/8
<x_jγ>
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Unfolded/Truth
Raw ratio
JES-corrected ratio
Data to MC JES (unfolded vs. truth, shape χ²) = 0.9380 ± 0.0040
p_T^γ [GeV]

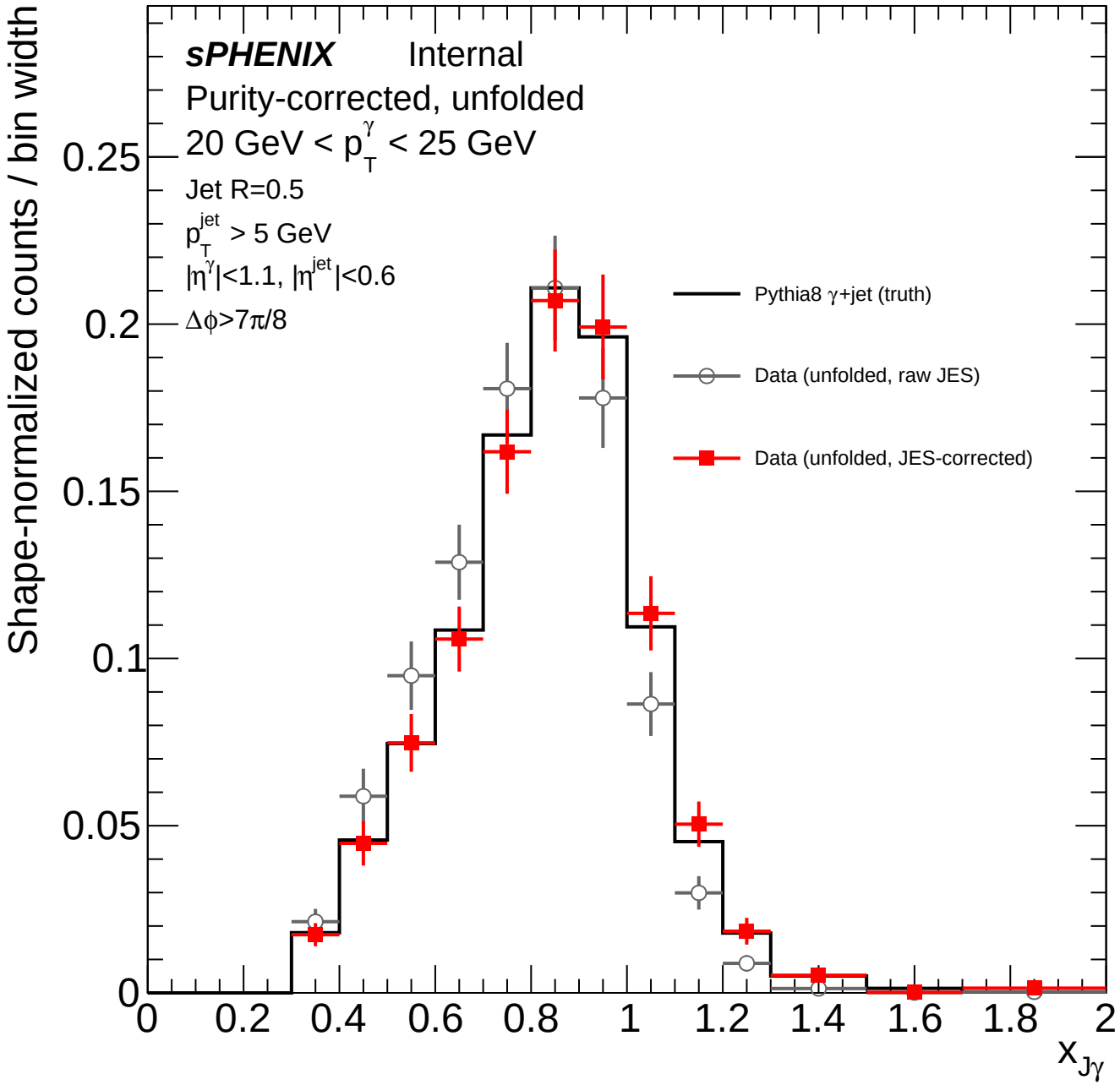

sPHENIX Internal
Purity-corrected, unfolded
15 GeV < p_T^γ < 20 GeV
Jet R=0.4
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.7
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

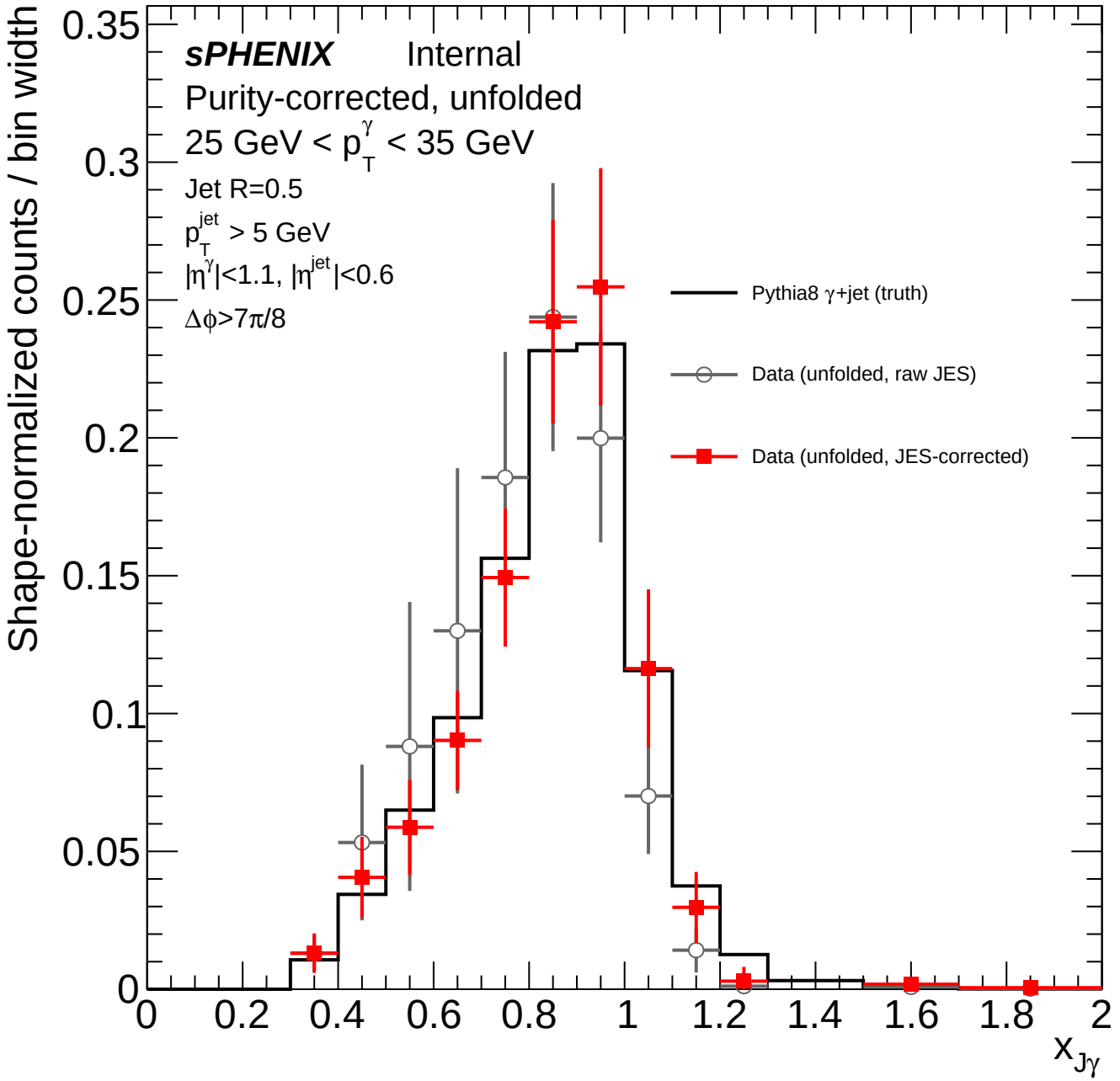

sPHENIX Internal
Purity-corrected, unfolded
20 GeV < p_T^γ < 25 GeV
Jet R=0.4
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.7
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
0
0.05
0.1
0.15
0.2
0.25
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
x_Jγ

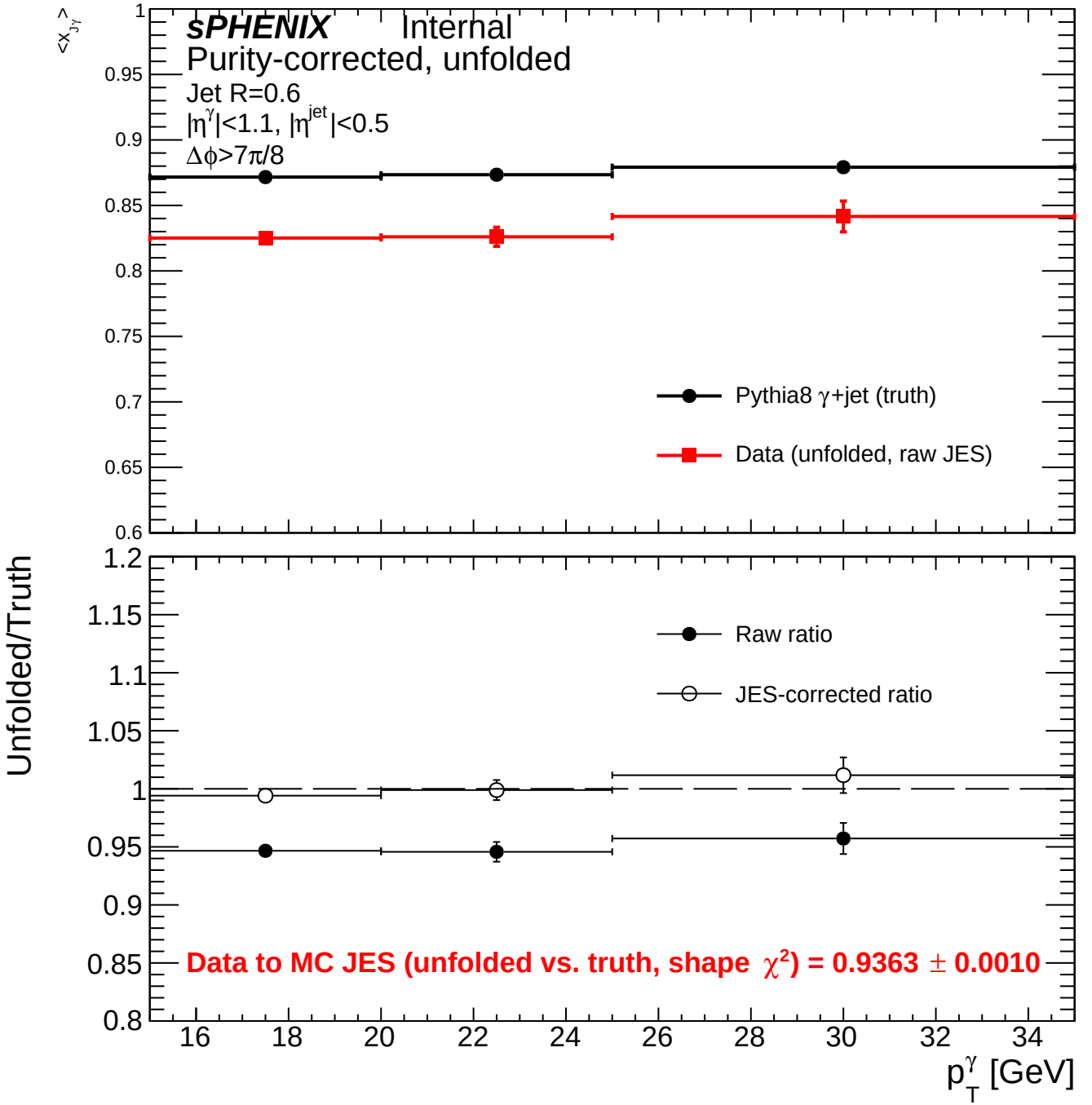

sPHENIX Internal
Purity-corrected, unfolded
25 GeV < p_T^γ < 35 GeV
Jet R=0.4
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.7
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.3
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

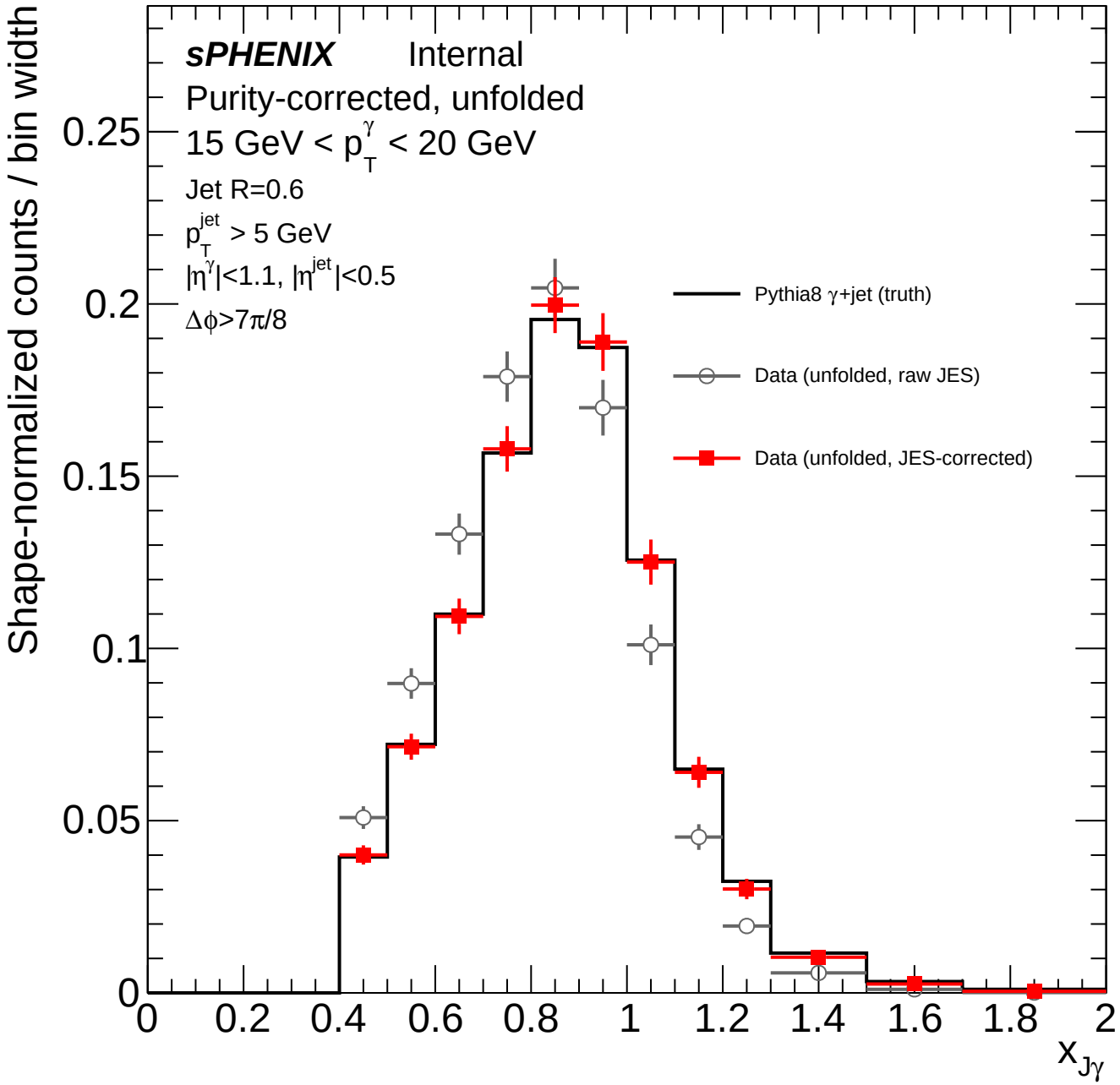

sPHENIX Internal
Purity-corrected, unfolded
Jet R=0.5
|η^γ|<1.1, |η^jet|<0.6
Δφ>7π/8
<x_jγ>
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Unfolded/Truth
Raw ratio
JES-corrected ratio
Data to MC JES (unfolded vs. truth, shape χ²) = 0.9338 ± 0.0017
p_T^γ [GeV]

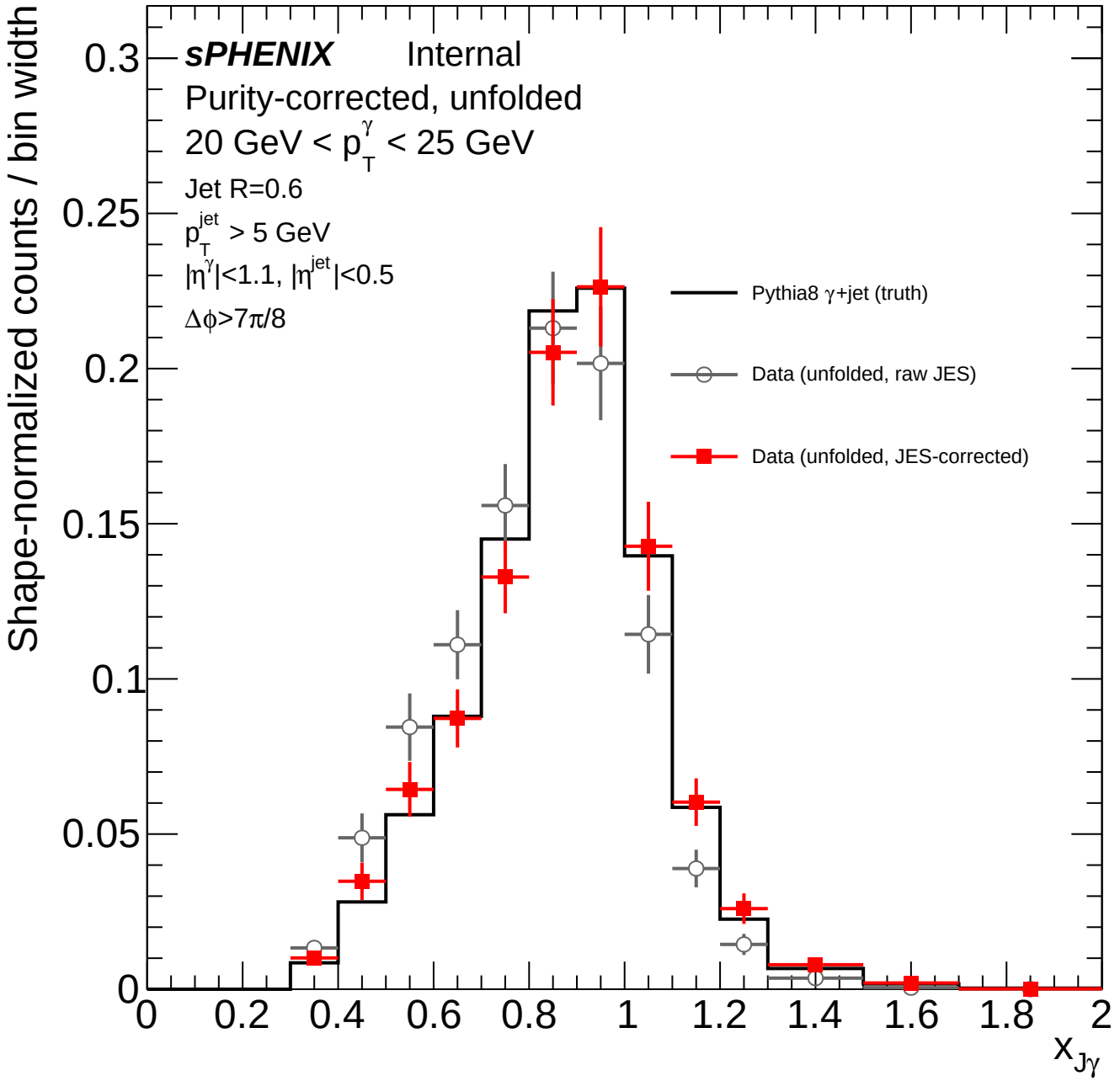

sPHENIX Internal
Purity-corrected, unfolded
15 GeV < $p_T^\gamma$ < 20 GeV
Jet R=0.5
$p_T^{jet}$ > 5 GeV
$|\eta^\gamma|<1.1$, $|\eta^{jet}|<0.6$
$\Delta\phi>7\pi/8$
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
$x_{J\gamma}$
0
0.05
0.1
0.15
0.2
0.25
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

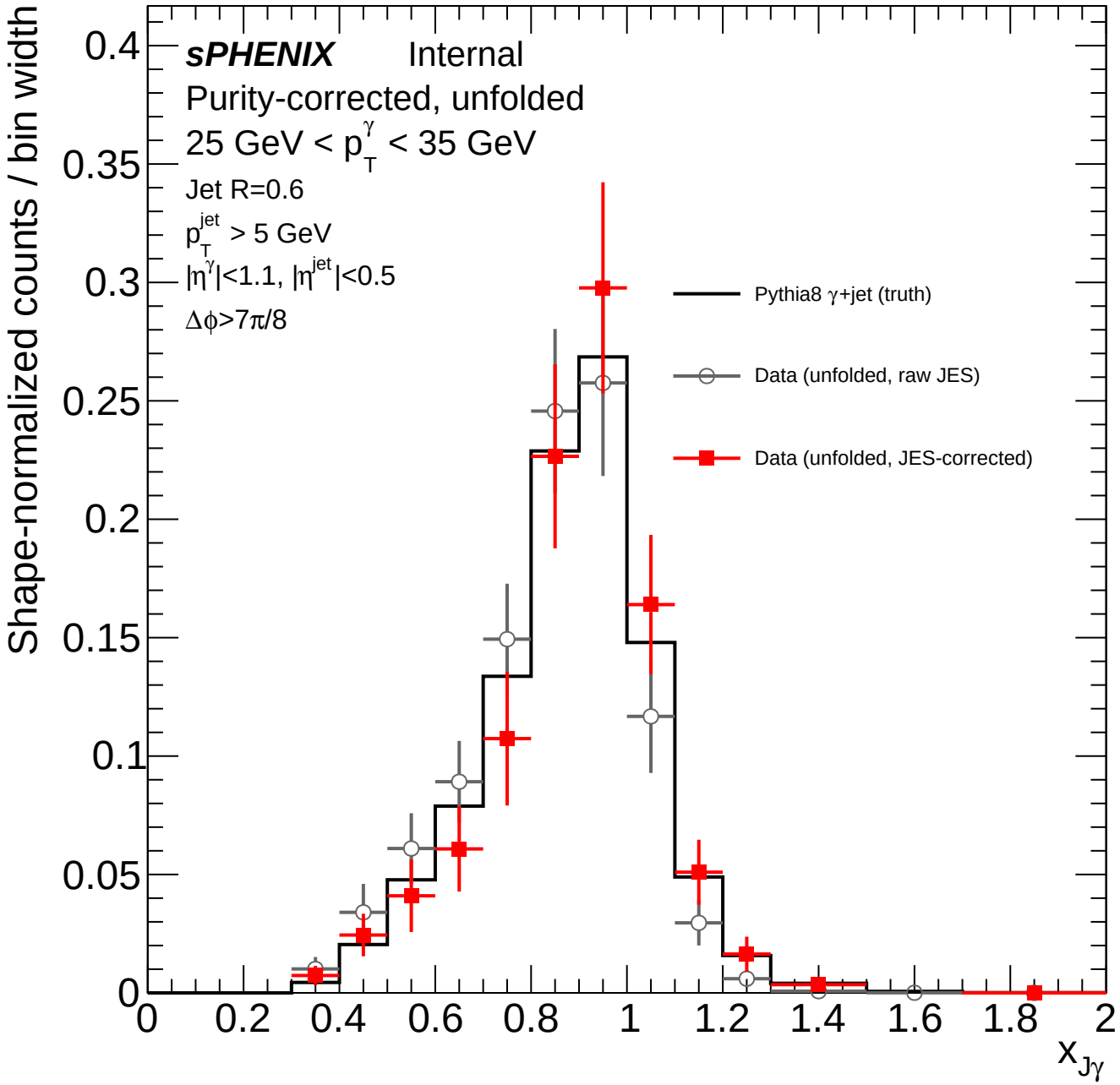

sPHENIX Internal
Purity-corrected, unfolded
20 GeV < p_T^γ < 25 GeV
Jet R=0.5
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.6
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
0
0.05
0.1
0.15
0.2
0.25
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
x_Jγ

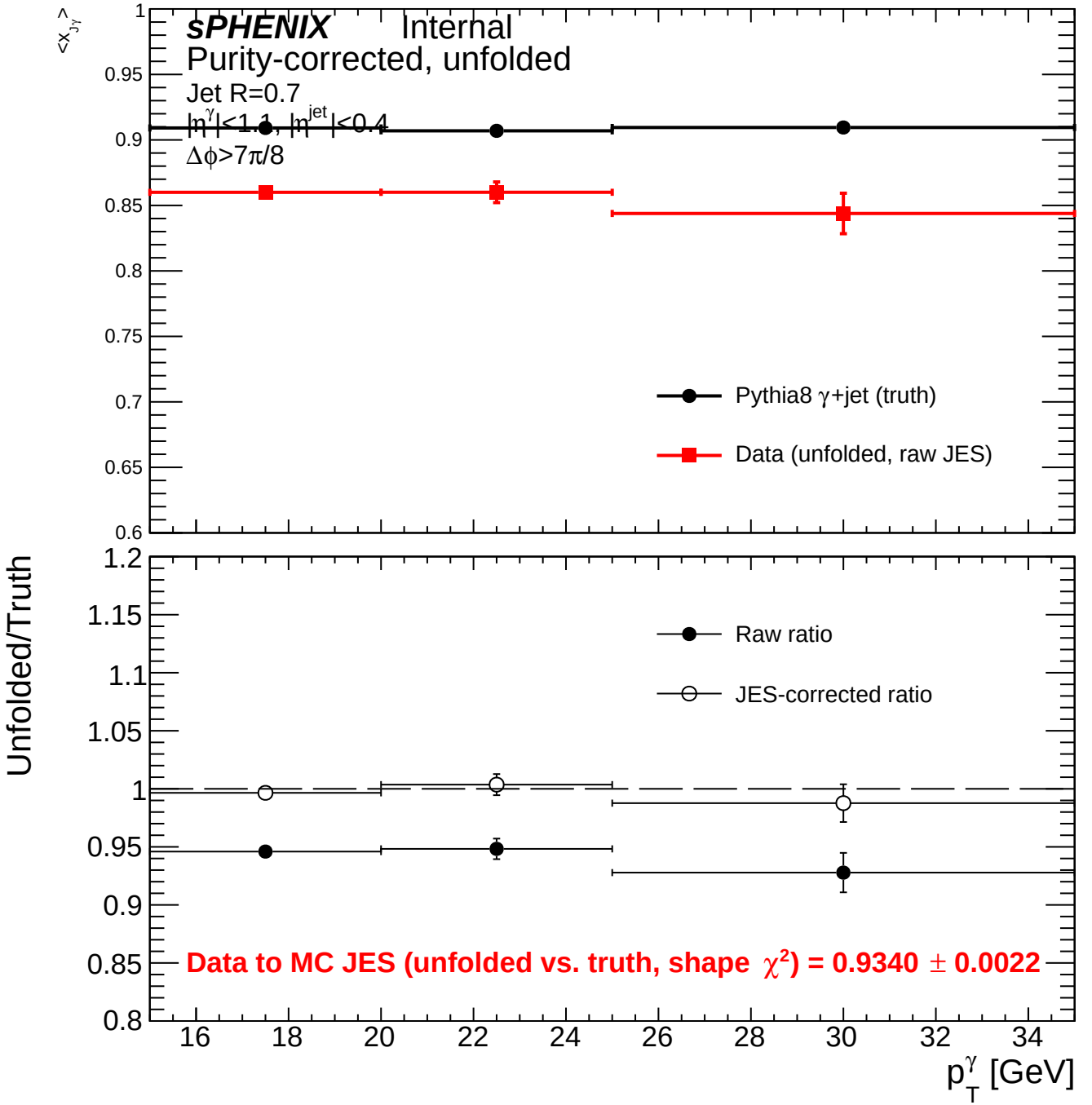

sPHENIX Internal
Purity-corrected, unfolded
25 GeV < $p_T^\gamma$ < 35 GeV
Jet R=0.5
$p_T^{jet}$ > 5 GeV
$|\eta^\gamma|$<1.1, $|\eta^{jet}|$<0.6
$\Delta\phi$>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
$x_{J\gamma}$
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

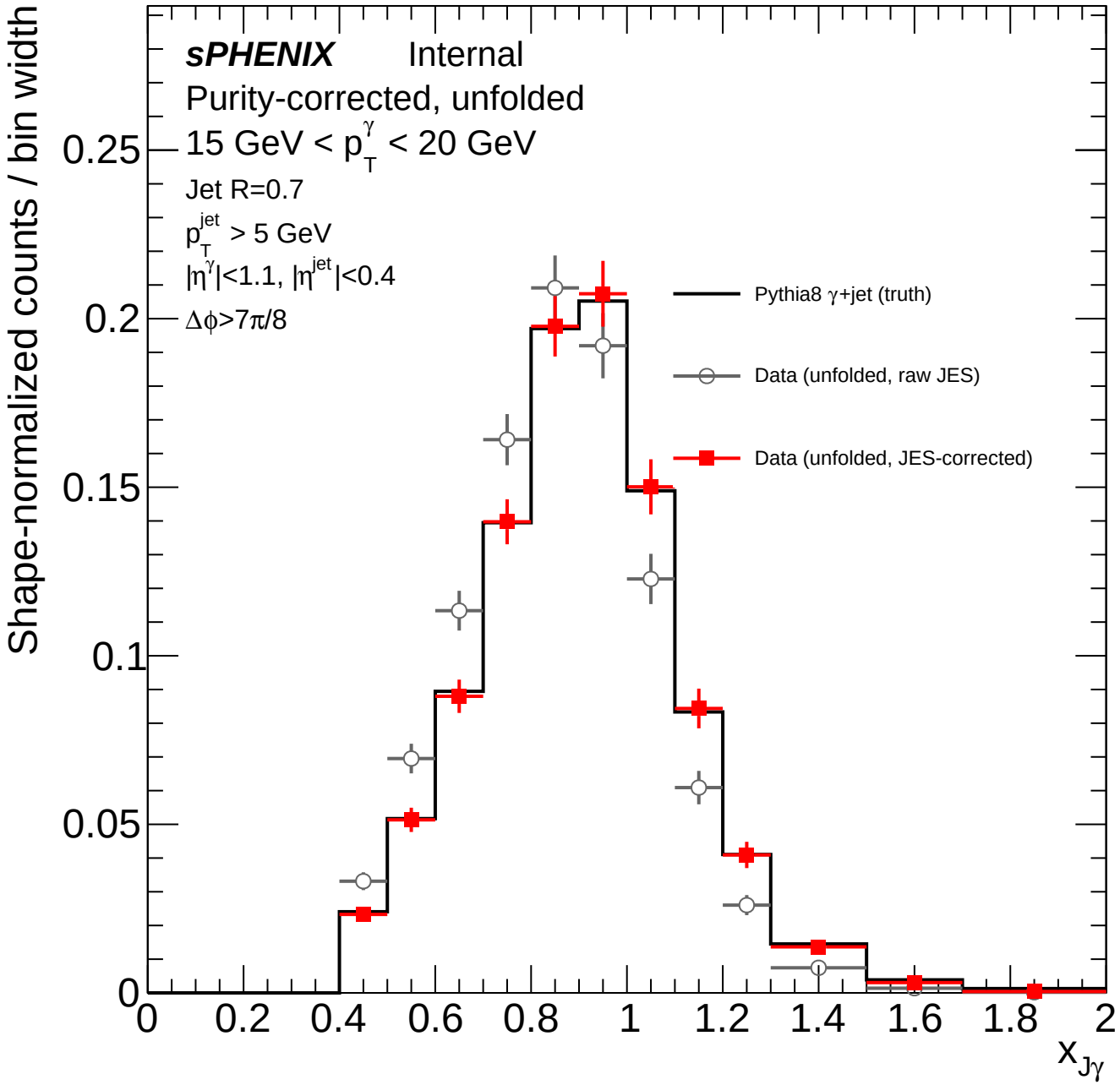

sPHENIX Internal
Purity-corrected, unfolded
Jet R=0.6
|η^γ|<1.1, |η^jet|<0.5
Δφ>7π/8
<x_jγ>
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Unfolded/Truth
Raw ratio
JES-corrected ratio
Data to MC JES (unfolded vs. truth, shape χ²) = 0.9363 ± 0.0010
p_T^γ [GeV]

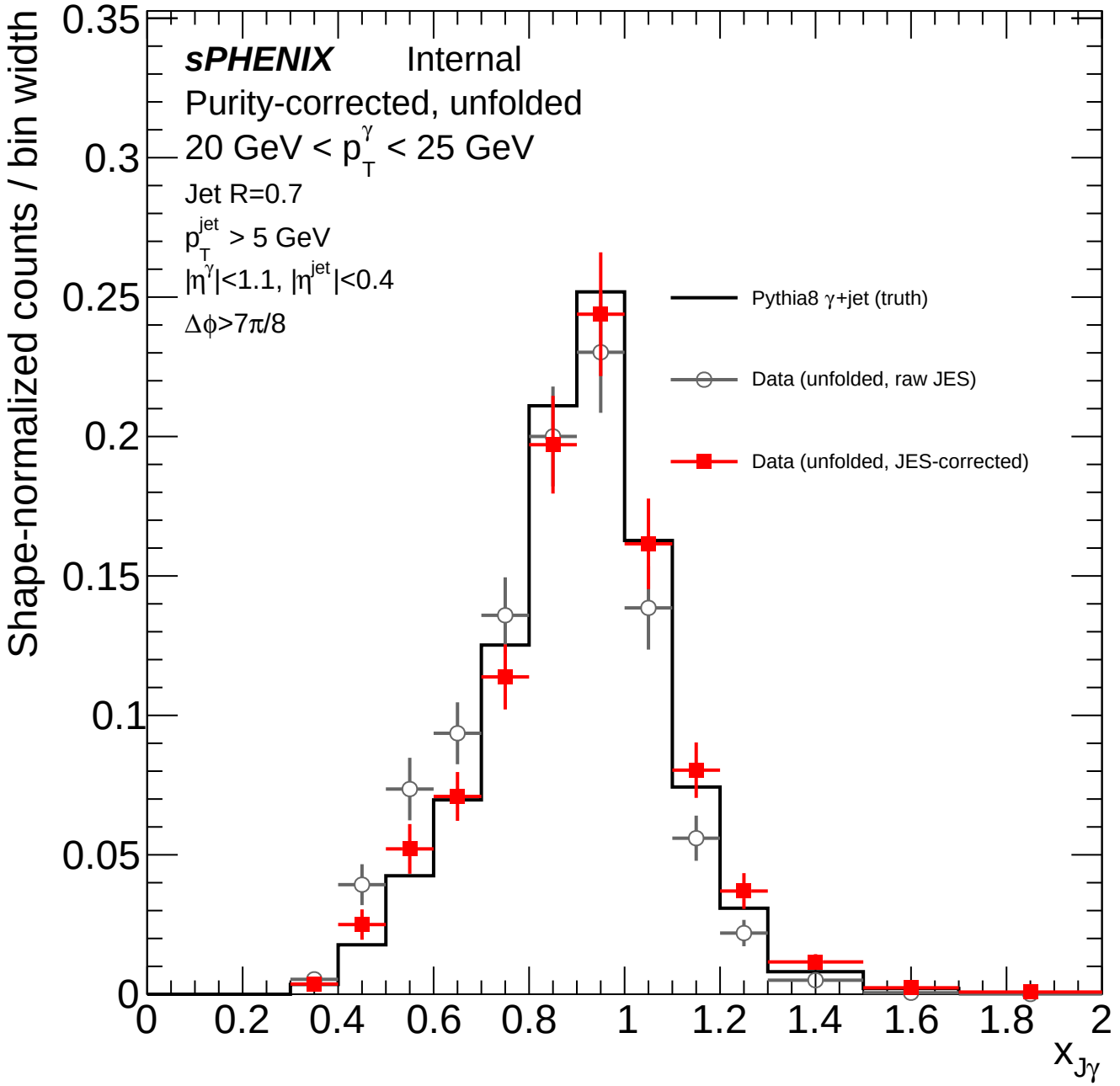

sPHENIX Internal
Purity-corrected, unfolded
15 GeV < p_T^γ < 20 GeV
Jet R=0.6
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.5
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
0
0.05
0.1
0.15
0.2
0.25
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
x_Jγ

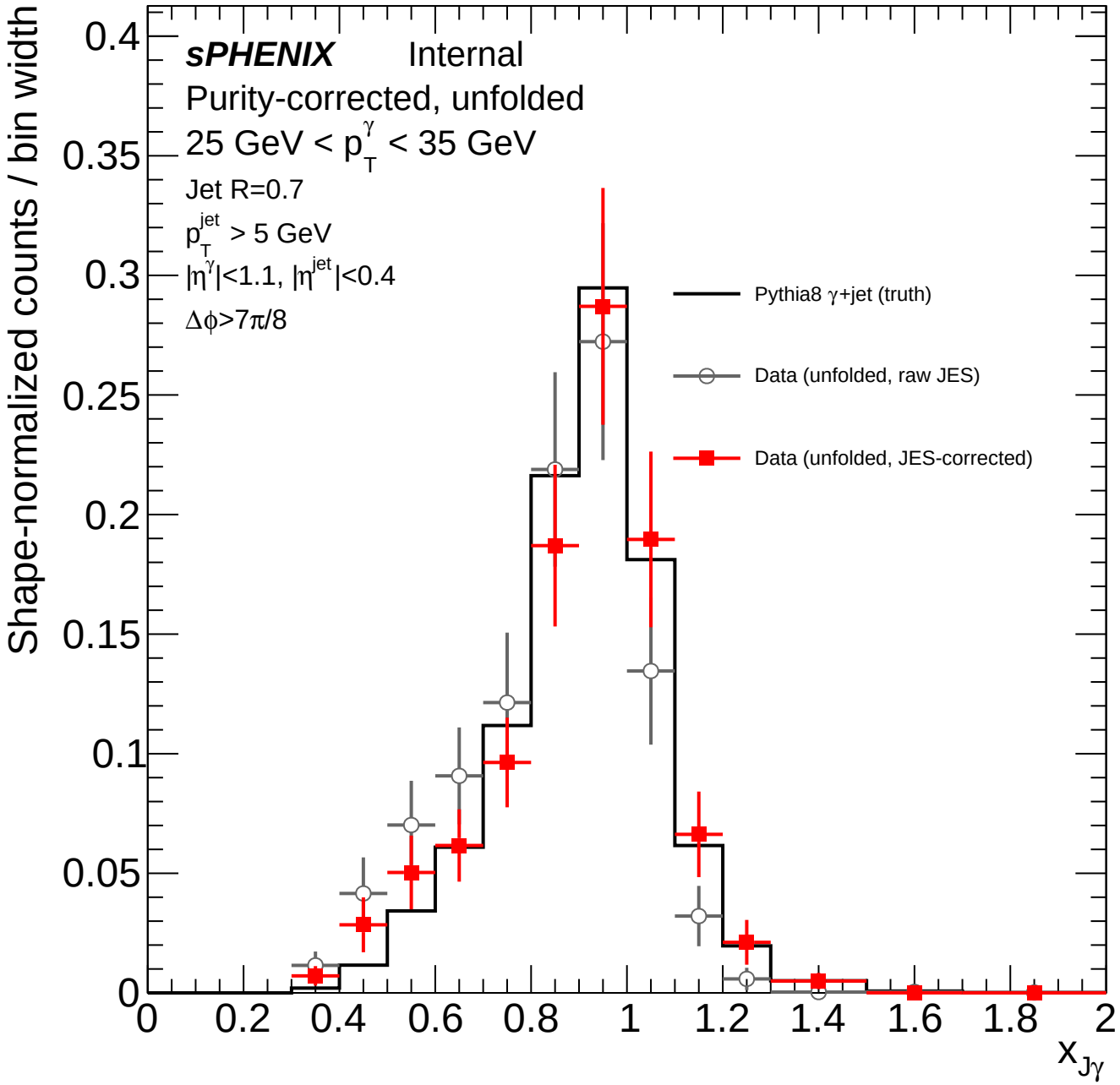

sPHENIX Internal
Purity-corrected, unfolded
20 GeV < p_T^γ < 25 GeV
Jet R=0.6
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.5
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.3
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

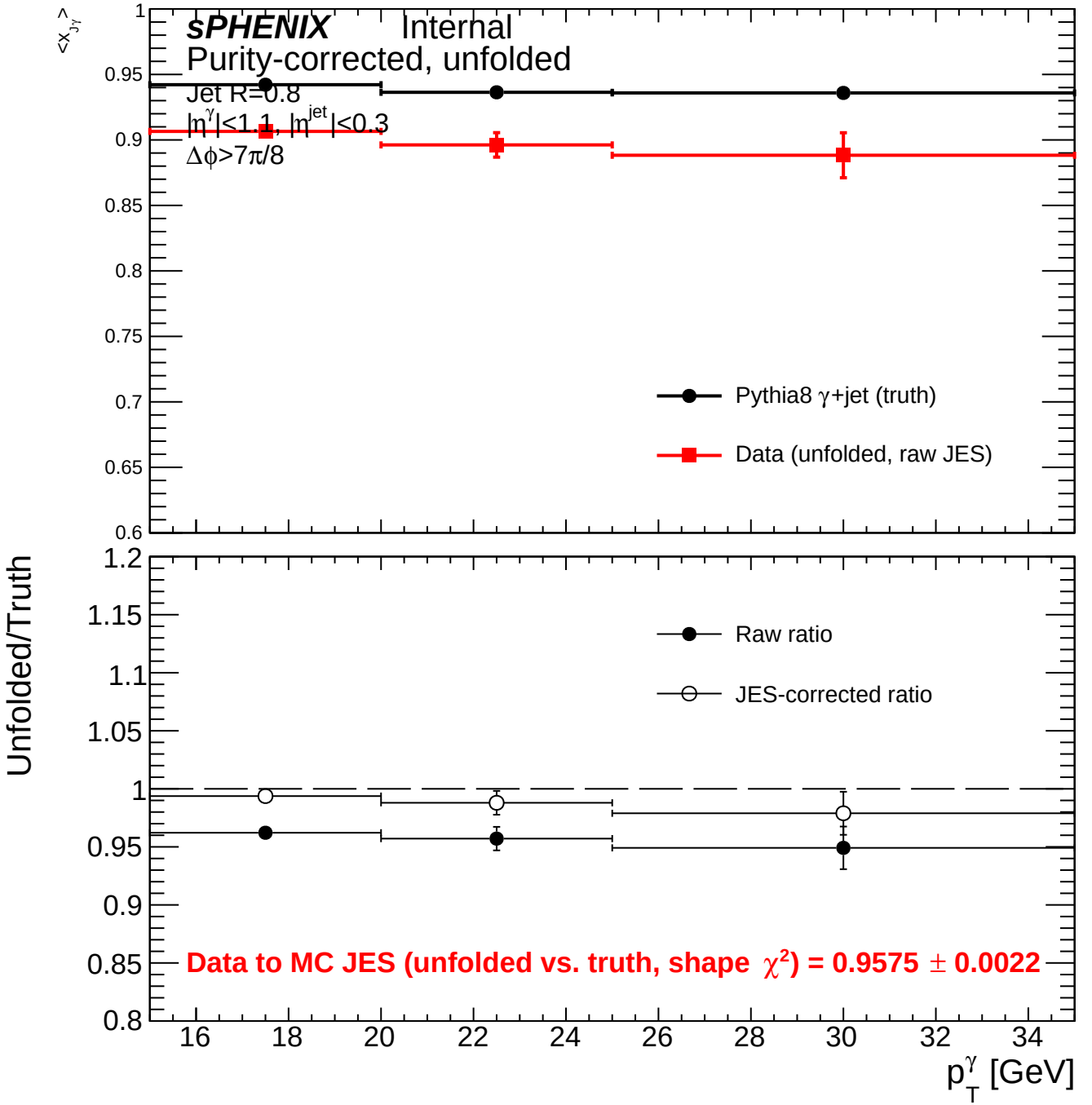

sPHENIX Internal
Purity-corrected, unfolded
25 GeV < $p_T^\gamma$ < 35 GeV
Jet R=0.6
$p_T^{jet}$ > 5 GeV
$|\eta^\gamma|<1.1$, $|\eta^{jet}|<0.5$
$\Delta\phi>7\pi/8$
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
$x_{J\gamma}$
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

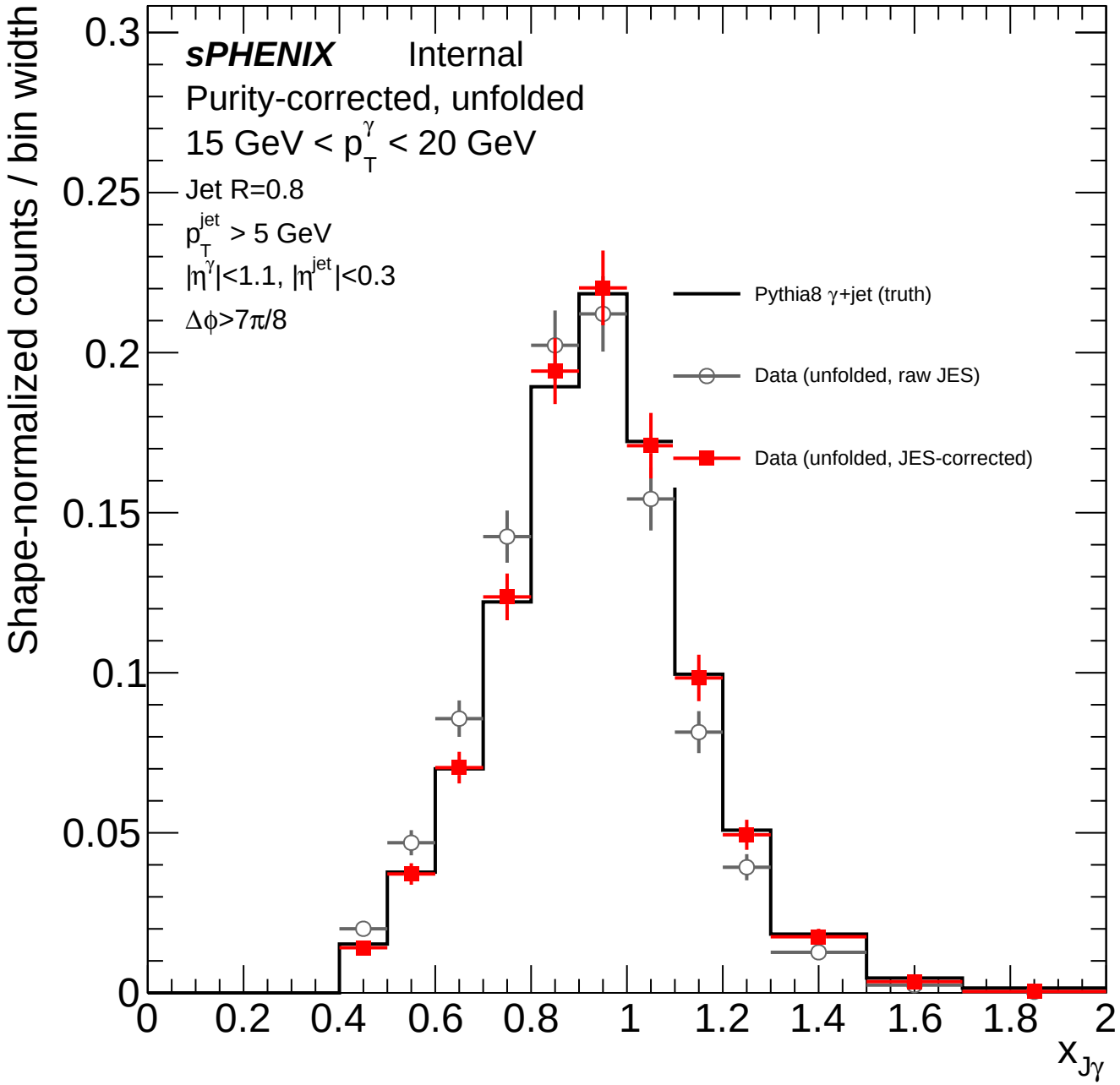

sPHENIX Internal
Purity-corrected, unfolded
Jet R=0.7
$|\eta^{\gamma}|<1.1$, $|\eta^{jet}|<0.4$
$\Delta\phi>7\pi/8$
$\langle x_{J\gamma} \rangle$
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Unfolded/Truth
Raw ratio
JES-corrected ratio
Data to MC JES (unfolded vs. truth, shape $\chi^2$) = 0.9340 ± 0.0022
$p_T^{\gamma}$ [GeV]

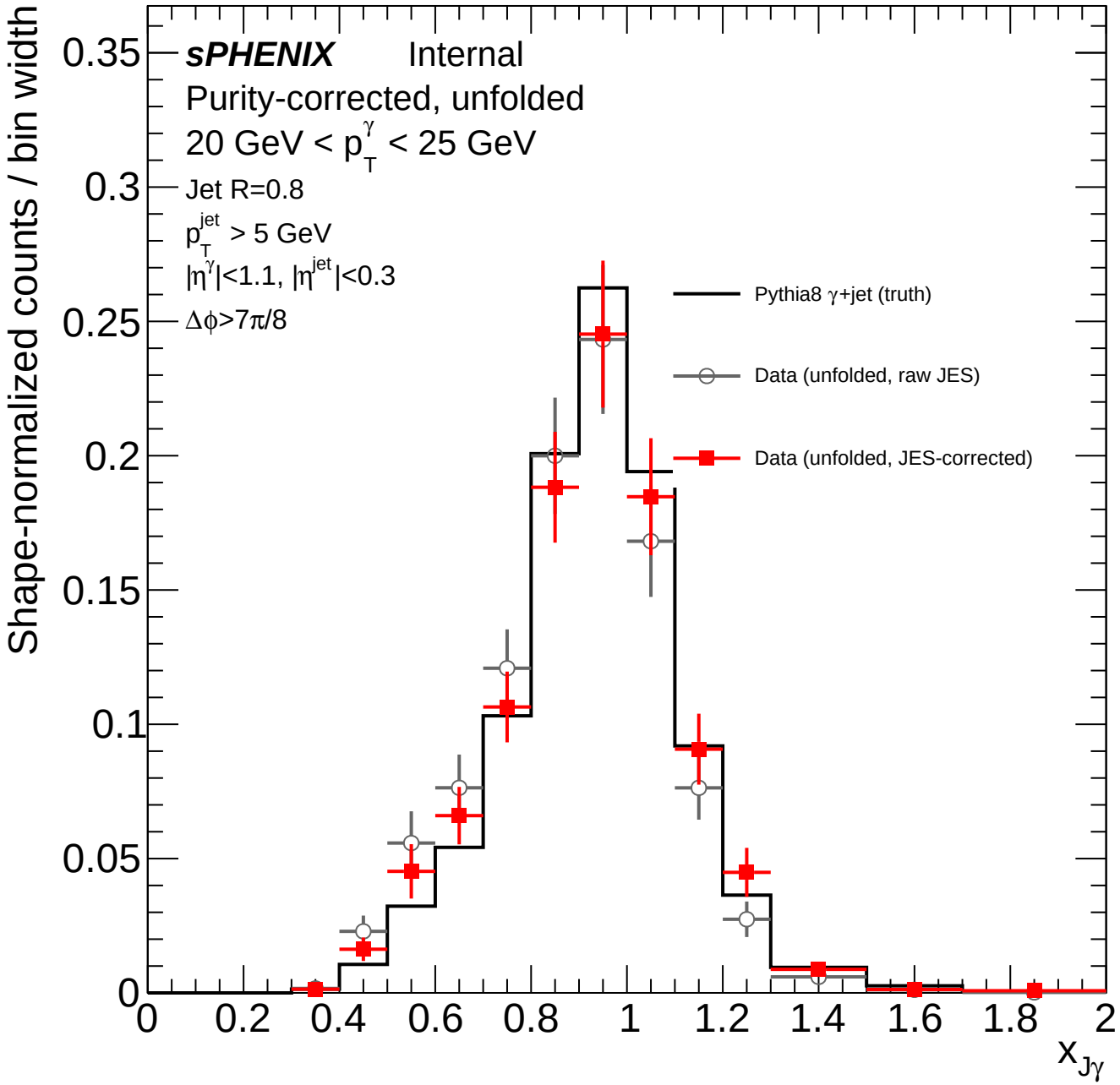

sPHENIX Internal
Purity-corrected, unfolded
15 GeV < $p_T^\gamma$ < 20 GeV
Jet R=0.7
$p_T^{jet}$ > 5 GeV
$|\eta^\gamma|$<1.1, $|\eta^{jet}|$<0.4
$\Delta\phi$>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
$x_{J\gamma}$
0
0.05
0.1
0.15
0.2
0.25
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

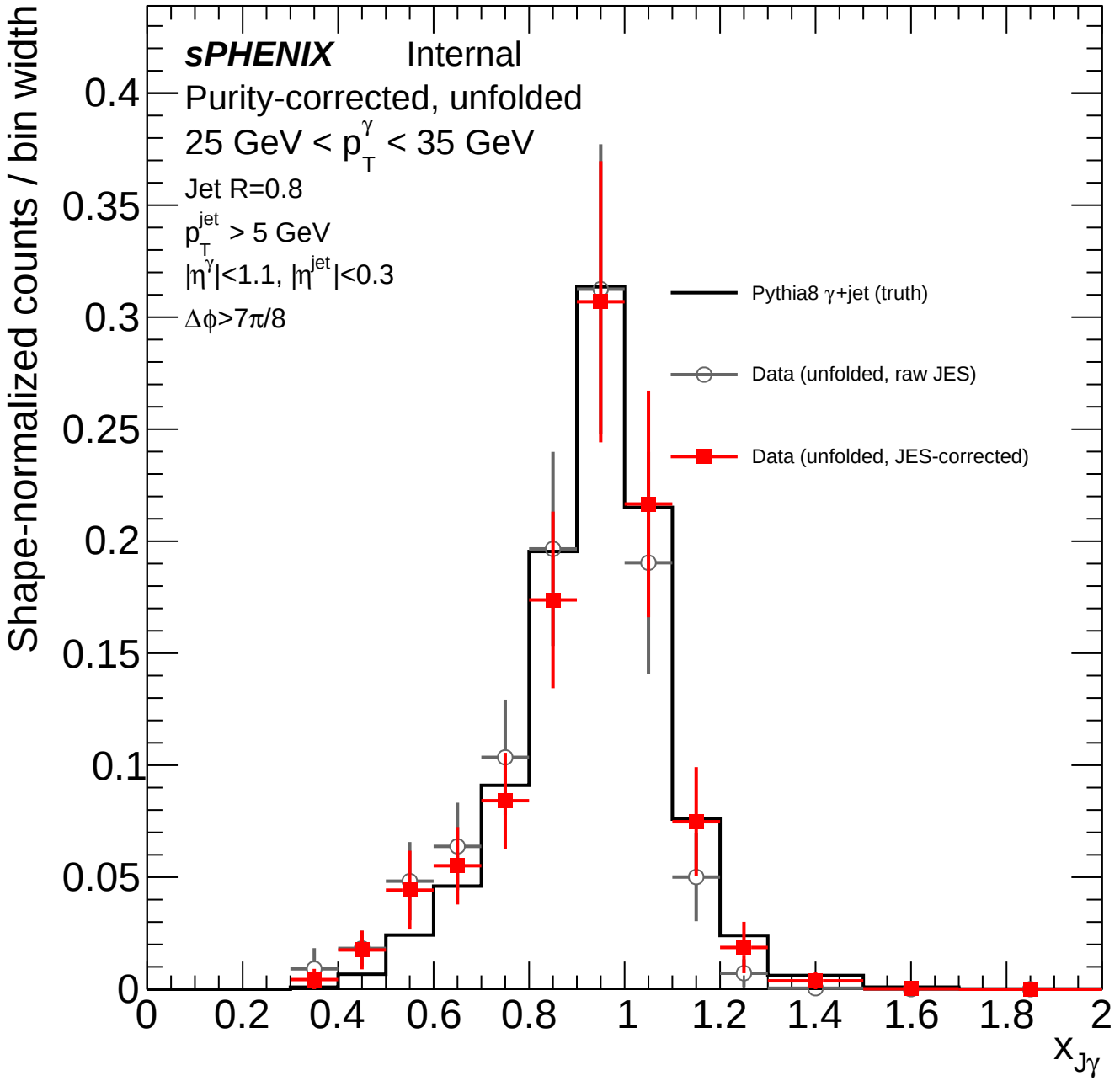

sPHENIX Internal
Purity-corrected, unfolded
20 GeV < p_T^γ < 25 GeV
Jet R=0.7
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.4
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
Purity-corrected, unfolded
25 GeV < p_T^γ < 35 GeV
Jet R=0.7
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.4
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
Purity-corrected, unfolded
Jet R=0.8
|η^γ|<1.1, |η^jet|<0.3
Δϕ>7π/8
<x_jγ>
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Unfolded/Truth
Raw ratio
JES-corrected ratio
Data to MC JES (unfolded vs. truth, shape χ²) = 0.9575 ± 0.0022
p_T^γ [GeV]


sPHENIX Internal
Purity-corrected, unfolded
15 GeV < p_T^γ < 20 GeV
Jet R=0.8
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.3
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.3
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
Purity-corrected, unfolded
20 GeV < p_T^γ < 25 GeV
Jet R=0.8
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.3
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
Purity-corrected, unfolded
25 GeV < p_T^γ < 35 GeV
Jet R=0.8
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.3
Δφ>7π/8
Pythia8 γ+jet (truth)
Data (unfolded, raw JES)
Data (unfolded, JES-corrected)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2